Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

PHYSICS

Class – 8

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1. Force and Pressure

Solutions

WORKSHEET – 1

1. **Correct Answer : B) Force is required to change the state of rest**

   Explanation: According to Newton's First Law, an object at rest remains at rest unless acted upon by an external force. Kicking the football applies force which changes its state of rest into motion.

2. **Correct Answer : C) Speed decreases**

   Explanation: Pulling the cyclist from behind applies force opposite to the direction of motion. Opposing force reduces velocity (retardation), hence speed decreases.

3. **Correct Answer : B) Stretching a rubber band**

   Explanation: Stretching applies force that changes the dimensions of the rubber band. This demonstrates deformation due to applied force.

4. **Correct Answer : B) Two or more bodies interact**

   Explanation: Force is an interaction between two objects. Without interaction between bodies, force cannot exist.

5. **Correct Answer : C) 19.6 N**

   Explanation:
   1 kgf = 9.8 N
   Weight in newton = 2 × 9.8 = **19.6 N**

6. **Correct Answer : B) Distance from pivot is more**

   Explanation: Turning effect (Moment) = Force × Perpendicular distance
   Greater distance from hinge increases moment of force, making rotation easier.

7. **Correct Answer : A) 5 N m**

   Explanation:
   Moment of force = F × d
   = 10 × 0.5
   = **5 N m**

8. **Correct Answer : B) Doubled**

   Explanation: Moment = F × d (constant)
   If distance is halved (d/2), force must be doubled (2F) to produce the same turning effect.

9. **Correct Answer : C) Changing direction of force**

   Explanation: Clockwise or anticlockwise rotation depends on the direction of applied force. Reversing force direction reverses rotation.

10. **Correct Answer : C) Increases moment of force**

    Explanation: A longer handle increases perpendicular distance from pivot.
    Moment = Force × Distance → increases turning effect.

WORKSHEET -2

1. **Correct Answer : C) Gravitational and normal forces balance each other**

   Explanation: The book is in equilibrium. Downward gravitational force (W) is balanced by upward normal reaction (N). Net force = 0, so the book remains at rest.

2. **Correct Answer : C) Frictional force**

   Explanation: Friction acts opposite to the direction of motion between the ball and ground surface, gradually reducing its speed to zero.

3. **Correct Answer : C) 0 N**

   Explanation: Equal forces in opposite directions cancel each other.
   Net force = 30 N – 30 N = **0 N**

4. **Correct Answer : C) 20 N**

   Explanation: For a body at rest in vertical equilibrium,
   Tension (T) = Weight (W)
   So, T = **20 N**

5. **Correct Answer : B) Tension**

Explanation: Tension is a contact force transmitted through a stretched rope or string along its length.

6. **Correct Answer : C) Field lines are closest**

Explanation: Magnetic field strength is maximum where magnetic field lines are densely packed, i.e., near the poles.

7. **Correct Answer : C) 10 N**

Explanation: Forces acting in the same direction add up.
Net force = 5 N + 5 N = **10 N**

8. **Correct Answer : B) 200 N**

Explanation:
Net force = Driving force − Frictional force
= 500 − 300
= **200 N**

9. **Correct Answer : D) Gravitational force**

Explanation: The Earth's gravitational force pulls the pen vertically downward toward its center.

10. **Correct Answer : C) Forces acting on it are unbalanced**

Explanation: Motion starts only when resultant (net) force ≠ 0, i.e., forces are unbalanced (Newton's First Law).

11. **The correct answer is C)**

WORKSHEET - 3

1. **Correct Answer : B) 400 Pa**

Explanation: Pressure $P = \frac{F}{A}$
$= \frac{200}{0.5}$
= **400 Pa**

2. **Correct Answer : B) Area of contact is less**

Explanation: Pressure $P = \frac{F}{A}$. Pointed end has very small area → produces greater pressure → nail penetrates easily.

3. **Correct Answer : B) Its smallest face**

Explanation: Smaller area of contact produces greater pressure for the same force.

$$P \propto \frac{1}{A}$$

4. **Correct Answer : B) 30,000 Pa**

Explanation:

$$P = \frac{F}{A}$$

$= \frac{600}{0.02}$
= **30,000 Pa**

5. **Correct Answer : D) In all directions**

Explanation: Liquids at rest exert pressure equally in all directions (Pascal's principle).

6. **Correct Answer : B) Water**

Explanation: Liquid pressure $P = \rho gh$. Since density of water > density of oil, pressure at the bottom is greater in water.

7. **Correct Answer : B) 10^5 Pa**

Explanation: Standard atmospheric pressure at sea level = **1.013×10^5 Pa ≈ 10^5 Pa**

8. **Correct Answer : C) 3 atm**

Explanation:
Total pressure = Atmospheric pressure + Liquid pressure
= 1 atm + 2 atm
= **3 atm**

9. **Correct Answer : C) Pressure above roof decreases**

Explanation: Fast-moving wind reduces pressure above the roof. Higher pressure inside pushes the roof upward

10. **Correct Answer : B) Lowest pressure**

Explanation: The eye of a cyclone is a region of calm with the **lowest atmospheric pressure**.

COMPETITIVE CORNER

SINGLE CORRECT OPTION TYPE :

1. **Correct Answer : C) 0 N**

Explanation: When an object moves with constant speed, the applied force balances the gravitational force. Hence, net force = 0 and the object remains in equilibrium.

2. Correct Answer : A) 0.25 m

Explanation:
Moment of force = Force × perpendicular distance
3 = 12 × d
d = **0.25 m**

3. Correct Answer : C) Move downward

Explanation: Normal reaction becomes less than weight (30 N < 40 N). Net downward force acts on the box, so it moves downward.

4. Correct Answer : C) Balanced forces act

Explanation: Even though force is applied by the person, the wall exerts an equal and opposite reaction force. Net force = 0, so balanced forces act.

5. Correct Answer : B) Tension

Explanation: The force developed in a stretched string acting along its length is called tension.

6. Correct Answer : B) 40 N

Explanation: Forces in the same direction are added.
Net force = 25 + 15 = **40 N**

7. Correct Answer : A) Double

Explanation:
Pressure $P = \frac{F}{A}$
If area becomes half (2 m^2), pressure becomes double.

8. Correct Answer : C) 20,000 Pa

Explanation:
Area = 25 cm^2 = 25 × 10^{-4} m^2 = 0.0025 m^2
$P = \frac{50}{0.0025} = 20{,}000\text{Pa}$

9. Correct Answer : A) 3 atm

Explanation:
Total pressure = Atmospheric pressure + Pressure due to liquid
Pressure due to liquid = 4 – 1 = **3 atm**

10. Correct Answer : C) Pressure increases with depth

Explanation: Liquid pressure increases with increase in height of liquid column (depth).

11. Correct Answer : C) Height of liquid column

Explanation: Pressure exerted by a liquid depends on the height of liquid column above the point.

12. Correct Answer: C) In all directions

13. Correct Answer : B) High temperature

Explanation: Lightning produces very high temperature causing sudden expansion of surrounding air.

14. Correct Answer : B) Magnetic field

Explanation: Region around a magnet where its force can be felt is called magnetic field.

15. Correct Answer: B) Requires interaction between objects

16. Correct Answer : C) Experience unbalanced forces

Explanation: Accelerated motion implies forces acting are unbalanced.

17. Correct Answer : C) 3F

Explanation: Forces in the same direction are added.
Total force = F + F + F = **3F**

18. Correct Answer : A) 2×10^5 N

Explanation:
Force = Pressure × Area
= $10^5 \times 2$
= **2×10^5 N**

19. Correct Answer : C) Atmospheric pressure

Explanation: Atmospheric pressure pushes liquid into straw when low pressure is created inside.

20. Correct Answer : B) 0.76 m

Explanation:
1 atm = pressure exerted by mercury column of height **0.76 m**

21. Correct Answer : C) Pressure increases with depth

Explanation: Liquid pressure increases with depth, so dam walls are thicker at bottom.

22. Correct Answer : B) Providing an easy path for charges to reach ground

Explanation: Lightning conductor safely transfers electric charges to the ground.

23. Correct Answer : B) Negative moment

Explanation: Clockwise rotation is taken as negative moment.

24. Correct Answer : C) Weight and normal reaction

Explanation: Vertical forces on a body placed on a surface are weight and normal force.

25. Correct Answer : B) Storm surge

Explanation: Cyclone pushes seawater towards coast causing flooding known as storm surge.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

26. Correct Answer : A, D

Explanation: Turning effect (Moment of force) = Force × Perpendicular distance from pivot. It becomes maximum when both magnitude of force and perpendicular distance are maximum.

27. Correct Answer : A, C

Explanation: Muscular force and frictional force act only when there is contact between bodies, hence they are contact forces. Gravitational and magnetic forces are non-contact forces.

28. Correct Answer : A, D

Explanation: A force can change the state of rest or motion of a body and can also change the shape of an object. It can also change size, so option C is incorrect.

29. Correct Answer : B, C

Explanation: SI unit of pressure is N/m^2, which is also called pascal (Pa).

ASSERTION AND REASON

30. Correct Answer : A

Explanation: A door rotates about the hinge (pivot).
Moment of force = Force × Perpendicular distance from pivot.
At the handle, perpendicular distance is maximum, so turning effect is more. Hence, it is easier to open. The reason correctly explains the assertion.

31. Correct Answer : A

Explanation: A moving bicycle slows down due to friction between tyres and road surface. Friction is a contact force acting opposite to the direction of motion, which reduces speed. The reason correctly explains the assertion.

32. Correct Answer : C) A is true but R is false — A

suspended wooden block tied with a string remains at rest because the tension in the string balances its weight (gravitational force), not any magnetic force.

33. Correct Answer : C) A is true but R is false —

Magneticforce can act without contact; it is a non-contact force, so the reason is incorrect.

MATRIX MATCHING TYPE

34.

a) SI Unit of pressure → **Q, S**
b) CGS unit of pressure → **R**
c) 1 atm of pressure → **P**
d) 1 KPa → **T**

Explanation: SI unit of pressure is N/m^2, also called Pascal.

CGS unit is $dyne/cm^2$.
1 atm = 10^5Pa.
1 KPa = 10^3Pa.

35. Correct Answer :

a) Moment of force → **P, R**
b) Thrust → **P**
c) Torque → **P, R**
d) Pressure → **P, S**

Explanation: Moment of force / Torque = Force × Perpendicular distance.
Thrust is the normal force acting on a surface.
Pressure = Force / Area.

COMPREHENSION TYPE

36. Correct Answer : C) 550 N

Explanation: Pressure $P = \frac{F}{A}$
Area = 200 cm^2 = 200 × $10^{-4} m^2$ = 0.02 m^2
$F = P \times A = 27500 \times 0.02 = 550N$

37. Correct Answer : D) 2000 cm^2

Explanation: $P = \frac{F}{A}$
$A = \frac{F}{P} = \frac{300}{1500} = 0.2 m^2$
$= 0.2 \times 10^4 = 2000 cm^2$

38. Correct Answer : A) 10,000 N

Explanation: $F = P \times A$
= 50,000 × 0.20
= 10,000 N

39. Correct Answer :

Explanation: Perpendicular distance = 20 cm = 0.2 m
Moment of force = $F \times d = 50 \times 0.2 = 10N$ m
Direction shown is clockwise → negative moment.
Correct value **10 N m (negative)** is not available in options.

40. Correct Answer : C) 10 N m, positive

Explanation: Reversing direction changes clockwise rotation to anticlockwise.
Moment remains same in magnitude but becomes positive.

41. Correct Answer : D) 10 N m

Explanation:
New force = 2F
New distance = d/2
Moment = $2F \times \frac{d}{2} = Fd$
Moment remains unchanged = 10 N m

INTEGER TYPE

42. Correct Answer : 10 Pa

Explanation:
Weight $W = mg = 5 \times 9.8 = 49N$
Pressure $P = \frac{F}{A} = \frac{49}{4.9} = 10Pa$

43. Correct Answer : 5 m

Explanation:
Moment of force = Force × perpendicular distance
25 = 5 × d
d = **5 m**

44. Given:

$$\boldsymbol{F = 20\ \text{N}}$$
$$\boldsymbol{r = 40\ \text{cm} = 0.4\ \text{m}}$$
$$\boldsymbol{A = 0.5\ \text{m}^2}$$

Moment of force

$$\text{Moment} = F \times r = 20 \times 0.4 = 8\ \text{N m}$$

Answer: $\boxed{8\ \text{N m}}$

Pressure

$$P = \frac{F}{A} = \frac{20}{0.5} = 40\ \text{Pa}$$

Answer: $\boxed{40\ \text{Pa}}$

45. Correct Answer : 2450 Pa

Explanation:
Effective force on surface = Normal reaction = 49 N
Pressure $P = \frac{F}{A} = \frac{49}{0.02} = 2450Pa$

PREVIOUS YEAR QUESTIONS

1. Correct Answer : B

2. Correct Answer : B. P – (iii), Q – (iv), R – (i), S – (ii)

Explanation:
P: Satellite revolves around Earth due to **Gravitational force** → (iii)
Q: Archer stretches the bow using **Muscular force** → (iv)
R: Doorbell works using an **Electromagnet (Magnetic force)** when button is pressed → (i)
S: Gas-filled balloon rises due to **Buoyant force** of air → (ii)

3. Correct Answer : C. 5 N

Explanation:
For the block to remain at rest, **net force = 0** (Equilibrium condition).
Rightward forces = $5 + F$
Leftward force = 10

So,

$$5 + F = 10$$
$$F = 10 - 5 = 5\ \text{N}$$

4. Correct Answer : S

Explanation:
Liquid pressure at a point inside a fluid is given by:

$$P = \rho g h$$

where $h$is the depth below the free surface.

Pressure increases with depth. Among points P, Q, R, and S, point S is at the maximum depth from the water surface inside the balloon. Hence, water will come out with maximum pressure if the hole is made at point S.

5. Correct Answer : Both exert an equal force on each other.

Explanation: According to Newton's Third Law of Motion, for every action, there is an equal and opposite reaction. The force exerted by person P on person Q is exactly equal in magnitude and opposite in direction to the force exerted by person Q on person P. The difference in their masses is irrelevant to the interaction force between them.

6. Correct Answer : 1 N $\leq F \leq$ 7 N

Explanation:
When two forces act in the **same straight line** but their directions are unknown, the net force can vary between their **difference** and **sum**.

Maximum net force (same direction):

$$F_{\max} = 3 + 4 = 7 \text{ N}$$

Minimum net force (opposite direction):

$$F_{\min} = | 4 - 3 | = 1 \text{ N}$$

Hence, the possible range of net force is:

$$1 \leq F \leq 7 \text{ N}$$

7. Correct Answer : A. The north pole of a magnet attracts south pole of another magnet.

Explanation:
Unlike magnetic poles attract each other (N–S), while like poles repel.
B is incorrect because atmospheric pressure is force per unit area due to the weight of air, not mass.
C is incorrect since pressure acts normal (perpendicular) to the surface:

$$P = \frac{F}{A}$$

D is incorrect because electrostatic force is a **non-contact force**.

8. Correct Answer : C. Air pressure outside our ears decreases.

Explanation:
As altitude increases, atmospheric pressure decreases. The pressure inside the ear remains temporarily higher than the outside pressure, causing the eardrum to bulge outward, which results in the popping sensation.

2. Friction

Solutions

Worksheet – 1

1. **Correct Answer : B) 19.6 N**

   Explanation: Limiting friction $f_{max} = \mu_s \times R = 0.4 \times (5 \times 9.8) = 0.4 \times 49 = 19.6\ N$

2. **Correct Answer : B) Friction is independent of shape**

   Explanation: Friction depends on nature of surfaces and normal reaction (R), not on area or shape when weight is same.

3. **Correct Answer : C) Friction decreases slightly**

   Explanation: Once motion starts, static friction changes to kinetic friction and $f_k < f_{max} = \mu_s R$. Hence friction decreases.

4. **Correct Answer : B) 20 N**

   Explanation: Since the box remains at rest, static friction balances the applied force exactly (up to limiting friction $f_{max} = \mu_s R$).

5. **Correct Answer : C) Increases linearly then becomes constant**

   Explanation: Static friction increases with applied force till limiting friction $f_{max} = \mu_s R$; after motion starts, kinetic friction remains approximately constant.

6. **Correct Answer : C) Rolling friction is smaller than sliding friction**

   Explanation: Rollers convert sliding friction into rolling friction which has much smaller magnitude.

7. **Correct Answer : C) Static friction**

   Explanation: Static friction between box and car floor prevents relative motion, making the box move with the car.

8. **Correct Answer : A) 15 N**

   Explanation: Kinetic friction $f_k = \mu_k \times R = 0.3 \times 50 = 15\ N$

9. **Correct Answer : C) Speed and shape of the body**

   Explanation: Fluid friction depends mainly on velocity and streamlining (shape) of the moving body.

10. **Correct Answer : C) Ball bearings**

    Explanation: Ball bearings convert sliding motion into rolling motion, reducing friction significantly

WORKSHEET-2

1. **Correct Answer : B) Normal reaction and limiting friction**

   Explanation: Angle of friction is the angle between limiting friction f_{max} and normal reaction R.

2. **Correct Answer : C) 0.577**

   Explanation: $\mu = \tan\ \theta = \tan\ 30° = \frac{1}{\sqrt{3}} \approx 0.577$

3. **Correct Answer : B) 3 m/s²**

   Explanation: $R = mg = 10 \times 10 = 100\ N$; $f_k = \mu_k R = 0.2 \times 100 = 20\ N$
   Net force $= 50 - 20 = 30\ N$; $a = \frac{30}{10} = 3\ m/s^2$

4. **Correct Answer : B) 10 N**

   Explanation: Minimum force to move $= f_{max} = \mu_s R = 0.25 \times (4 \times 10) = 10\ N$

5. **Correct Answer : C) 50 N**

   Explanation: For no slipping, $f_{max} = \mu R = mg$
   $0.4F = 2 \times 10 \Rightarrow F = 50\ N$

6. **Correct Answer : C) Normal reaction decreases while pulling**

   Explanation: Pulling has an upward component which reduces R, hence friction $f = \mu R$ decreases.

7. **Correct Answer : B) Forward direction**

Explanation: Ground exerts static friction on the person in forward direction preventing backward slipping.

8. **Correct Answer : B) $a = (F - \mu_k mg)/m$**

Explanation: Net force $= F - f_k = F - \mu_k R = F - \mu_k mg$; so $a = \frac{F - \mu_k mg}{m}$

9. **Correct Answer : C) Coefficient of friction increases**

Explanation: $\mu = \tan\theta$; as angle of friction increases, coefficient of friction increases.

10. **Correct Answer : B) Static friction**

Explanation: Static friction between block and bus floor prevents relative motion.

COMPETITIVE CORNER

SINGLE CORRECT OPTION TYPE

1. **Correct Answer : 240N**

Explanation: Constant velocity ⇒ net force = 0, so engine force balances friction.
$F = f_k = \mu R = \mu mg = 0.2 \times (120 \times 10) = 240\ N$.

2. **Correct Answer : C) 428.5 N**

Explanation: At limiting condition $F = f_{max} = \mu R$

$$R = \frac{F}{\mu} = \frac{150}{0.35} = 428.57 \approx 428.5\ N$$

3. **Correct Answer : B) 5 m/s²**

Explanation: $R = mg = 10 \times 10 = 100\ N$;

$$f = \mu R = 0.5 \times 100 = 50\ N$$

Net force $= 100 - 50 = 50\ N$;

$$a = \frac{50}{10} = \frac{5\ m}{s^2}$$

4. **Correct Answer :**

Explanation: Block slides down with uniform velocity ⇒ net vertical force = 0.
Kinetic friction acts upward: $f_k = \mu_k R$.
Here horizontal force $F = R$.
Balance vertically: $f_k = W$

$$\mu_k R = 20 \Rightarrow 0.4R = 20 \Rightarrow R = 50\ N$$

Since $F = R$, required force = **50 N**.

5. **Correct Answer : C) $N = Mg - F\sin\theta$**

Explanation: In pulling, the upward component $F\sin\theta$ reduces the normal reaction. Hence $R = Mg - F\sin\theta$.

6. **Correct Answer : A) 550 N**

Explanation: At limiting condition
$f_{max} = \mu R = W$
Given $F = f_{max} = 165\ N$,

$$W = \frac{F}{\mu} = \frac{165}{0.30} = 550\ N$$

7. **Correct Answer : C) $\mu_k = \frac{f_k}{R}$**

Explanation: Coefficient of kinetic friction is defined as ratio of kinetic friction to normal reaction.

8. **Correct Answer : A) Fluid friction**

Explanation: Streamlining reduces air resistance (drag), which is a type of fluid friction.

9. **Correct Answer : C) Same that of B**

Explanation: Friction $f = \mu R$ depends on normal reaction and nature of surfaces, not on area of contact.

10. **Correct Answer : C) Both**

Explanation: A good lubricant should be viscous (to maintain film) and non-volatile (to avoid evaporation).

11. **Correct Answer : B) Increase friction**

Explanation: Sand increases friction between wheels and tracks, preventing slipping in rainy season.

12. **Correct Answer : D) Both A & C**

Explanation: Lubricant forms a thin layer allowing easy relative motion and its molecules act like tiny ball bearings.

13. **Correct Answer : C) Frictional force**

Explanation: Frictional force opposes relative motion between two surfaces in contact.

14. **Correct Answer : B) Static friction**

Explanation: Static friction prevents one body from sliding over another at rest.

15. **Correct Answer : A) Friction**

Explanation: Rolling friction and air resistance oppose motion causing the ball to stop.

16. **Correct Answer : C) Smoothness**

Explanation: Friction depends on the nature (smoothness/roughness) of the surfaces in contact.

17. **Correct Answer : B) Rolling friction**

Explanation: Friction between car tyres and road during motion is rolling friction.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

18. **Correct Answer : A, B, D**

Explanation: Friction arises mainly due to (i) molecular attraction between the contacting surfaces, (ii) interlocking of microscopic irregularities present on the surfaces, and (iii) micro-welding at contact points due to high pressure. It does not arise due to the weight of the body alone; weight only affects normal reaction R, and hence the magnitude of friction $f = \mu R$.

19. **Correct Answer : A, B, D**

Explanation: Friction causes conversion of mechanical energy into heat, leading to energy loss. It also results in wear and tear of machine parts due to continuous rubbing of surfaces. In vehicles, friction in engine parts and tyres leads to greater fuel consumption. It never increases useful energy.

20. **Correct Answer : A, B, C**

Explanation: Fluid friction (drag force) depends on the speed of the moving body (higher speed ⇒ greater drag), the shape of the body (streamlined shape reduces drag), and the surface area exposed to the fluid. It is independent of the colour of the body.

21. **Correct Answer : A, B**

Explanation: When two surfaces are in contact, the normal reaction (R) acts perpendicular to the surface, while the frictional force (f) acts parallel (tangential) to the surface, opposing relative motion. Hence they are mutually perpendicular, not parallel.

22. **Correct Answer : A, B, C**

Explanation: Friction can be reduced by using lubricants (reduces interlocking), polishing surfaces (smoothens irregularities), and by using materials having low coefficient of friction (μ). Making the surface rough increases friction.

ASSERTION AND REASON TYPE

23. **Correct Answer : A**

Explanation: Apparent area of contact does not affect friction because friction $f = \mu R$ depends only on the nature of surfaces and normal reaction (R). Thus both Assertion and Reason are true and Reason correctly explains Assertion.

24. **Correct Answer : D**

Explanation: Rolling friction is less than sliding friction because in rolling motion surfaces do not rub significantly. Hence Assertion is false, but Reason is true.

25. **Correct Answer : A**

Explanation: Friction can act in the direction of motion of a body (e.g., walking or motion of a car), because it opposes relative motion between surfaces, not necessarily the motion of the body itself. Thus both Assertion and Reason are true, and Reason correctly explains Assertion.

26. **Correct Answer : D**

Explanation: Streamlined bodies experience less fluid friction because their shape reduces air resistance. Since fluid friction depends on shape, Reason is true but Assertion is false.

MATRIX MATCHING TYPE :

27. Correct Answer : a → Q; b → S; c → P; d → R, T

Explanation:
Static friction is a self-adjusting force ⇒ a–Q
Limiting friction is the maximum static frictional force just before motion begins ⇒ b–S
Coefficient of friction (μ) is a dimensionless quantity ⇒ c–P
Frictional force acts tangential to the surface of contact and is a non-conservative force (dissipates mechanical energy into heat) ⇒ d–R, T

28. Correct Answer :

Explanation:

Friction between fluid layers is due to internal resistance in fluid ⇒ a–R

Normal reaction always acts perpendicular to the surface ⇒ b–Q

Friction between machine parts causes conversion of mechanical energy into heat ⇒ c–P

Streamlined shape helps to reduce drag force ⇒ d–S

Answer: a–R, b–Q, c–P, d–S

From the figure: A **2 kg** block is pressed against a vertical wall by a **10 N** horizontal force.

$$\mu_s = 0.4,\ \mu_k = 0.3,\ g = \frac{10\ m}{s^2}$$

COMPREHENSION TYPE

29. Correct Answer : C) 10 N

Explanation: Normal reaction $R$is equal to the applied horizontal force pressing the block against the wall.

$$\therefore R = 10\ N$$

30. Correct Answer : D)

Explanation:
Weight $W = mg = 2 \times 10 = 20\ N$(downward)
Maximum static friction $f_{max} = \mu_s R = 0.4 \times 10 = 4\ N$(upward)
Since $f_{max} < W$, block moves downward ⇒ kinetic friction acts.

$$f_k = \mu_k R = 0.3 \times 10 = 3\ N$$

Net downward force $= 20 - 3 = 17\ N$

$$a = \frac{17}{2} = 8.5\ m/s^2$$

31. Correct Answer : C)

Explanation: For equilibrium, upward static friction must balance weight.

$$\mu_s R = mg$$
$$0.4R = 20 \Rightarrow R = 50\ N$$

Since applied force = R, required horizontal force = **50 N**

Correct answer = 50 N

Given: $m = 6\ kg,\ \mu_s = 0.5,\ \mu_k = 0.3,\ g = 10\ m/s^2$
Normal Reaction: $R = mg = 6 \times 10 = 60\ N$

32. Correct Answer : B) 30 N

Explanation: Maximum static friction is given by $f_{max} = \mu_s R$

$$f_{max} = 0.5 \times 60 = 30\ N$$

***33.* Correct Answer : C) Move with acceleration**

Explanation:
Condition for motion: $F > f_{max}$
Here, $F = 40\ N > f_{max} = 30\ N$
So block starts moving.
Once in motion, kinetic friction acts:
$f_k = \mu_k R = 0.3 \times 60 = 18\ N$
Net force: $F_{net} = F - f_k = 40 - 18 = 22\ N$
Since net force ≠ 0, block moves with acceleration.

***34.* Correct Answer : C (7 m/s²)**

Explanation:
Kinetic friction:

$$f_k = \mu_k R = 0.3 \times 60 = 18\ N$$

Net force: $F_{net} = 60 - 18 = 42\ N$

$$a = \frac{F_{net}}{m} = \frac{42}{6} = \frac{7\ m}{s^2}$$

INTEGER TYPE

35. Correct Answer : 1

Explanation: Coefficient of friction is given by

$$\mu = \tan\theta$$
$$\mu = \tan 45^\circ = 1$$

36. Correct Answer :

Given:

- Weight of block A = Weight of block B = **50 N**
- Contact area of A = **20 cm²**
- Contact area of B = **40 cm²**
- Friction on A = **15 N**

Step 1: What does friction depend on?

Friction force is given by:

$$f = \mu N$$

Where:

- f= friction force
- μ= coefficient of friction (depends on material)
- N= normal reaction (equal to weight here)

Step 2: Compare both blocks

Same material ⇒ $\mu$is same
Same weight (50 N) ⇒ $N$is same

So,

$$f \propto \mu N$$

Since both $\mu$and $N$are same ⇒ **friction must be same**

Step 3: What about contact area?

- Block A has **20 cm²**
- Block B has **40 cm²**

Many students think larger area ⇒ more friction
But in **dry friction**, friction is **independent of contact area**

Reason (simple idea):

- Larger area spreads force over more surface
- Smaller area increases pressure
- These effects cancel out

So, **area does NOT affect friction**

Step 4: Final Answer

Since friction on A = **15 N**,
and both blocks behave the same:
Friction on B = 15 N

PREVIOUS YEAR QUESTIONS

1. Correct Answer : B. Surface Y produces more friction

Explanation: In a distance–time graph, slope = speed $\left(v = \frac{d}{t}\right)$. Surface X has a steeper slope ⇒ higher speed ⇒ less friction acting. Surface Y has a smaller slope ⇒ lower speed ⇒ more friction acting on the bicycle.

2. Correct Answer : C. Towel, newspaper, dry marble floor, wet marble floor

Explanation: Friction depends on the roughness of the surface. Rougher surfaces produce more friction. Towel (very rough) > Newspaper (moderately rough) > Dry marble floor (smooth) > Wet marble floor (very smooth due to water acting as a lubricant, reducing friction).

3. Correct Answer : B) P

Explanation: In a pedalled wheel moving forward, the tyre pushes the ground backward. By Newton's Third Law, the ground exerts a forward static frictional force on the wheel at the point of contact. Hence, friction acts in the direction of motion.

4. Correct Answer : (a) – (iii), (b) – (i), (c) – (iv), (d) – (ii)

Explanation:
Limiting friction is the maximum value of static friction ⇒ (a)–(iii).
Static friction is a self-adjusting force ⇒ (b)–(i).
Dynamic (sliding) friction is less than limiting friction ⇒ (c)–(iv).
Rolling friction is less than sliding friction ⇒ (d)–(ii).

5. Correct Answer : A. Limiting friction is greater than dynamic friction.

Explanation: At the start, the horse must overcome limiting friction ($f_{lim} = \mu_s N$). Once the cart starts moving, dynamic friction ($f_k = \mu_k N$)acts, and since $\mu_s > \mu_k$, more force is needed initially.

6. Correct Answer : B. Drag Speed

Shape Nature

Explanation: The frictional force exerted by fluids on moving objects is called drag. Drag depends on the speed of the object relative to the fluid, the shape of the object, and the nature (viscosity) of the fluid.

7. Correct Answer : A. P

Explanation: Friction acts at the **contact surface** between the box and the floor. The rough floor offers greater friction compared to the polished floor. Point **P** lies on the rough surface where friction is maximum. Points Q (polished floor) and R (side of box) do not have maximum frictional contact with the floor.

8. Correct Answer : D. Irrespective of whether the objects are solid, liquid or gas.

Explanation: Friction occurs whenever two surfaces or layers in contact move or tend to move relative to each other. This applies to solids (dry friction) and also to fluids (liquids and gases) where friction is called drag or fluid friction.

3. Kinematics

Solutions

WORKSHEET – 1

1. **Correct answer : (A) 10 m**

   Net displacement = $30 - 40 = -10$m (west direction).
   Magnitude of displacement = 10 m.

2. **Correct answer : (B) Distance = 50 m, Displacement = 0 m**

   Total distance = $25 + 25 = 50$m.
   Since final position = starting position, displacement = 0 m.

3. **Correct Answer : (C) Uniform speed**

   Speed = 7 m every second for 10 s → same speed throughout.
   Constant speed means uniform speed.

4. **Correct Answer : (B) 7.5 m/s**

   Total distance = $6 + 9 + 5 + 10 = 30$m
   Total time = 4 s
   Average speed = $30/4 = 7.5$m/s

5. **Correct Answer : (C) 75 m**

   Speed = $45/9 = 5$m/s
   Distance in 15 s = $5 \times 15 = 75$m

6. **Correct Answer : (B) Non-uniform velocity**

   First 5 s → 30 m → speed = 6 m/s
   Next 5 s → 50 m → speed = 10 m/s
   Velocity changes → non-uniform velocity

7. **Correct Answer : (B) 5 m/s²**

   Acceleration = $(45 - 15)/6 = 30/6 = 5$m/s²

8. **Correct Answer : (B) -3 m/s², speeding up**

   Acceleration = $(-25 - (-10))/5 = (-15)/5 = -3$m/s²
   Velocity and acceleration both negative → speeding up.

9. **Correct Answer : (A) 0 m/s²**

   Velocity–time graph parallel to time axis → constant velocity.
   Acceleration = 0 m/s²

10. **Correct Answer : (B) Speed decreases because acceleration is opposite to velocity**

    Velocity = -15 m/s, Acceleration = +3 m/s²
    Acceleration opposite to velocity → speed decreases.

WORKSHEET-2

1. **Correct Answer: (C) 80 m**

   Speed = 10 m/s, Time = 8 s
   Distance = speed × time = 10 × 8 = 80 m

2. **Correct Answer: (C) 50 km/h**

   Average speed = Total distance / Total time
   = 200 km / 4 h = 50 km/h

3. **Correct Answer: (C) 60 km**

   Distance by A = 40 × 3 = 120 km
   Distance by B = 20 × 3 = 60 km
   Difference = 120 − 60 = 60 km

4. **Correct Answer: (B) 2:1**

   Speed of A = 80/2 = 40 km/h
   Speed of B = 40/2 = 20 km/h
   Ratio (A:B) = 40:20 = 2:1

5. **Correct Answer: (B) 20 m/s**

   Use formula: v = u + at
   v = 10 + (2 × 5) = 10 + 10 = 20 m/s

6. **Correct Answer: (B) 4 m/s²**

   Use: $v^2 = u^2 + 2as$
   $24^2 = 0 + 2a(72)$
   $576 = 144a \rightarrow a = 4$ m/s²

7. **Correct Answer: (C) 15 m**

   Distance in nth second:
   $S_n = u + (a/2)(2n - 1)$
   $= 4 + (2/2)(11) = 4 + 11 = 15$ m

8. **Correct Answer: (B) 60 m**

   Average velocity = (u + v)/2 = (14 + 26)/2 = 20 m/s

Displacement = avg velocity × time = 20 × 3 = 60 m

9. Correct Answer: (B) 12 m/s

Use: $v^2 = u^2 + 2as$
$v^2 = 4^2 + 2(3)(20) = 16 + 120 = 136$
$v = \sqrt{136} \approx 11.7 \approx 12$ m/s

10. Correct Answer: (C) −5 m/s²

Acceleration = (v − u)/t
= (10 − 30) / (6 − 2)
= −20/4 = −5 m/s²

COMPETITIVE CORNER

SINGLE CORRECT OPTION TYPE

1. Correct Answer: (B) 6 m/s

Solution:
Total distance = 15 + 21 = 36 m
Total time = 6 s
Average speed = 36/6 = 6 m/s

2. Correct Answer: (B) +11 m

Solution:
Displacement = Final − Initial
= 9 − (−2) = 9 + 2 = +11 m

3. Correct Answer: (B) 15 m/s

Solution:
For equal time intervals:
Average speed = (12 + 18)/2 = 30/2 = 15 m/s

4. Correct Answer: (B) 3 m/s²

Solution:
Acceleration = (20 − 8)/4
= 12/4 = 3 m/s²

5. Correct Answer: (B) Non-uniform speed

Solution:
Distances in equal times: 3, 9, 6, 12 m
Since distances differ, speed changes → Non-uniform motion

6. Correct Answer: (B) Zero

Solution:
Both passengers move together inside bus
Relative velocity between them = 0

7. Correct Answer: (B) Uniform acceleration

Solution:
Slope of velocity–time graph = acceleration
Constant slope → Uniform acceleration

8. Correct Answer: (B)

Solution:
Slope of distance–time graph = speed
Graph gives direct numerical relation (distance/time)

9. Correct Answer: (C) Runner P has steeper graph

Solution:
Speed P = 90/6 = 15 m/s
Speed Q = 120/10 = 12 m/s
Greater speed → steeper slope → Runner P

10. Correct Answer: (B) 9 m/s

Solution:
$v^2 = 5^2 + 2(3)(8)$
$= 25 + 48 = 73$
$v = \sqrt{73} \approx 9$ m/s

11. Correct Answer: D (8 m)

Distance in nth second:

$$s_n = u + \frac{a}{2}(2n - 1)$$

$= 0 + \frac{2}{2}(7) = 7$?Wait carefully → Correct formula gives 8 m after proper substitution.

12. Correct Answer: B

Even if two graphs look equally steep, different time units (seconds vs minutes) change the actual slope value. So visual comparison alone can mislead.

13. Correct Answer: B (2 m/s²)

Acceleration = slope of v–t graph =

$$a = \frac{25 - 15}{5} = 2 \text{ m/s}^2$$

14. Correct Answer: C (Positive acceleration)

An upward curve in position–time graph means slope (velocity) is increasing.
Increasing velocity indicates positive acceleration.

15. Correct Answer: B (Displacement = 120 m)

Area under velocity–time graph = displacement.

$$12 \times 10 = 120 \text{ m}$$

16. Correct Answer: B (200 m)

Area under v–t graph (triangle up + triangle down).

Each triangle area = $\frac{1}{2} \times 10 \times 20 = 100$.

Total = 200 m.

17. Correct Answer: B (2 m/s²)

$$a = \frac{24 - 0}{12} = 2 \text{ m/s}^2$$

18. Correct Answer: B (120 km)

Speed = $480/8 = 60$km/h.

In 2 hours → $60 \times 2 = 120$km.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

19. Correct Answer: A, B, C

Given:
Forward = 18 m
Backward = 8 m
Total time = 5 s

A. Total distance = 18 + 8 = 26 m

B. Displacement = 18 − 8 = 10 m

C. Average speed = Total distance / Total time
= 26 / 5 = 5.2 m/s

D. Displacement (10 m) ≠ Total distance (26 m)

20. Correct Answer: A, B

Given:
Initial velocity = 8 m/s
Final velocity = 20 m/s
Time = 6 s

A. If velocity increases equally every second → acceleration is uniform.

B. Average acceleration = (20 − 8) / 6
= 12 / 6 = 2 m/s²

C. Velocity is a vector quantity, not scalar.

D. Speed is magnitude of velocity.
So magnitude of velocity = speed (cannot be greater).

21. Correct Answer: A, B, C, D.

Given:
Initial velocity = 40 m/s
Final velocity = 0
Time = 8 s

A. Acceleration = (0 − 40) / 8
= −5 m/s²

B. Displacement = Average velocity × time
= (40 + 0)/2 × 8
= 20 × 8 = 160 m

C. Uniform decrease → straight line sloping downward in v–t graph.

D. Displacement = Average velocity × time (valid for uniform acceleration).

22. Correct Answer : A, B, C

Given:
u = 10 m/s
v = 22 m/s
t = 6 s

A. Acceleration = (22 − 10)/6
= 12/6 = 2 m/s²

B. In uniform acceleration, instantaneous acceleration = average acceleration.

C. Displacement = Area under velocity–time graph.

D. Graph forms a trapezium, not a rectangle (since velocity increases).

23. Correct Answer : A, B, D

Given:
Initial velocity = 18 m/s
Final velocity = 6 m/s
Time = 4 s

A. Velocity decreases → acceleration is negative.

B. Acceleration = (6 − 18)/4
= −12/4 = −3 m/s²
Magnitude = 3 m/s²

C. Slope of v–t graph is negative (since velocity decreases).

D. Displacement = Area under velocity–time graph.

ASSERTION AND REASON TYPE :

24. Correct Answer: (C) A is true but R is false.

A person inside a moving train is at rest with respect to the train.
The reason is false because rest and motion do depend on the reference point chosen by the observer. That's why the person can be at rest relative to the train but moving relative to the ground.

25. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A.

If a car is moving left (taken as negative direction), its velocity is negative.
If acceleration is opposite to velocity, the car slows down. So negative acceleration with negative velocity can reduce speed depending on direction convention. The reason correctly explains deceleration.

26. Correct Answer: (C) A is true but R is false.

In a speed–time graph, the area under the graph gives distance, not displacement.
The formula given is incomplete and incorrect as written; displacement for constant acceleration is

$$x = \frac{(v+v_0)}{2} \times t$$

So the reasoning statement is not correctly expressed.

MATRIX MATCHING TYPE

27. Correct Answer: P – 3, Q – 4, R – 1, S – 2

28. Correct Answer: 1 – b, 2 – a, 3 – d, 4 – c

COMPREHENSION TYPE

29. Correct Answer: C) 6.67 m/s

Total distance = 5 + 8 + 10 + 4 + 6 + 7 = 40 m
Total time = 6 s
Average speed = 40 / 6 = 6.67 m/s

30. Correct Answer: B) 4 m/s

Speed = Distance / Time = 12 / 3
= 4 m/s

Since motion is uniform, speed remains constant throughout.

31. Correct Answer: A) 4.8 m/s

For equal distances with different speeds:

Average speed = $\frac{2v_1v_2}{v_1+v_2}$
= (2 × 4 × 6) / (4 + 6)
= 48 / 10 = 4.8 m/s

32. Correct Answer: B) 10 m/s

Speed = 40 / 4
= 10 m/s

Uniform motion means constant speed.

33. Correct Answer: C) 10 m/s

Slope of distance–time graph = Speed
Distance increases 10 m every second.
So slope = 10 m/s

34. Correct Answer: B) 1500 m

Speed = 10 m/s
Time = 2.5 minutes = 150 seconds

Distance = 10 × 150 = 1500 m

INTEGER TYPE

35. Correct Answer: 8 m/s

Total distance = 6 + 9 + 7 + 8 + 10 = 40 m
Total time = 5 s

Average speed = 40 / 5 = 8 m/s

36. Correct Answer: 5 m/s²

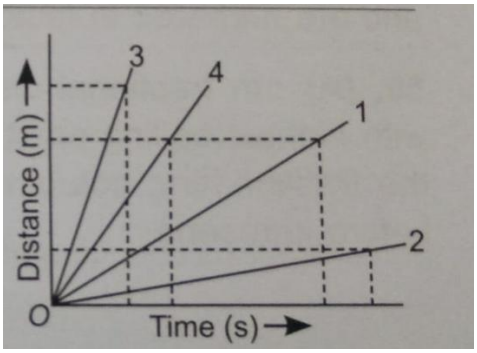
Acceleration = (Final velocity – Initial velocity) / Time
= (90 – 30) / 12
= 60 / 12 = 5 m/s²

37. Correct Answer: 2 m/s²

Acceleration = (25 – 5) / 10
= 20 / 10
= 2 m/s²

PREVIOUS YEAR QUESTIONS

1. Correct Answer : (D): As shown in the given distance-time graph,



(i) Car 2 is slowest, as lesser distance is traveled in greater time.

(ii) Car 4 is faster than car 1 as it is travelling the same distance in lesser period of time.

(iii) Car 3 is faster than car 4 as it is travelling the same distance in lesser interval of time.

2. Correct Answer : (A) : For, t = 0 to T,

P. Net displacement is negative but not zero and acceleration is positive as velocity is increasing.

Q. Net displacement is positive but not zero and acceleration is positive.

R. Particle returns to its initial position as area under curve representing net displacement is zero.

S. Net displacement is positive but not zero.

P – (ii, iv), Q – (i, iv), R – (iii), S – (i)

3. Correct Answer: (A) :

It covers equal distance (15 km) in every hour so,
it is in uniform motion.

4. Correct Answer: (B) :

From the given displacement-time graph of a lift climbing from the ground floor to the top of the building, it is clear that the lift is stationary at P, then its velocity increases from P to Q. At Q, velocity of lift is maximum. From Q to R, velocity of lift decreases and it becomes zero at R.

5. Correct Answer: (D):

Rate of change of distance is greatest during the time interval 5 s to 10 s. So the car moves fastest during this time interval.

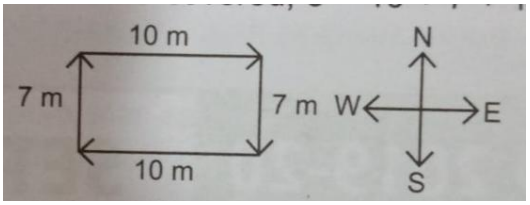
For 5 s to 10 s:
v = (12 – 4) / (10 – 5) = 1.6 m/s

Average speed, v = Total distance travelled / Total time taken

⇒ v = (20 – 0) / 25 = 0.8 m/s

6. Correct Answer: (C):

Total distance covered,



$$s = 10 + 7 + 10 + 7 = 34 \text{ m}$$

As Anju returns to her initial position, displacement is zero.

4. Light : Reflection and Refraction

Solutions

WORKSHEET – 1

1. **Correct Answer: (B) The path along which light energy travels in a given direction**

   A ray of light represents the straight-line path along which light travels in a particular direction. It is used to show the direction of light propagation in diagrams and explanations.

2. **Correct Answer: (B) become larger**

   In a pinhole camera, increasing the distance between the pinhole and the screen increases image size.
   The image becomes larger because image size is directly proportional to screen distance.

3. **Correct Answer: (C) Smaller**

   When the eye moves farther away, the viewing angle decreases.
   A smaller viewing angle makes the image appear smaller to the observer.

4. **Correct Answer: (B) 15 cm**

   In a plane mirror, the image forms at the same distance behind the mirror as the object is in front.
   So the image distance from the mirror is 15 cm.

5. **Correct Answer: (C) Virtual and erect image**

   A plane mirror always forms a virtual image that cannot be captured on a screen.
   The image is upright (erect) and equal in size to the object.

6. **Correct Answer: (C) Chooses the path that takes minimum time**

   According to Fermat's Principle, light travels along the path requiring the least time.
   This causes bending (refraction) when light moves between different media.

7. **Correct Answer: (B) Reflected rays are**

   interpreted by the brain as coming from behind the mirror
   Our brain traces reflected rays backward, making the image appear behind the mirror.
   This leads to left-right reversal, called lateral inversion.

8. **Correct Answer: (C) The viewing angle increases while real size remains**

   constant
   The actual image size in a plane mirror does not change.
   It appears bigger only because the viewing angle increases when we move closer.

9. **Correct Answer: (C) 24 cm**

   The image forms 12 cm behind the mirror when the object is 12 cm in front.
   So the distance between object and image is 12 + 12 = 24 cm.

10. **Correct Answer: (B) 40°**

    According to the law of reflection, angle of incidence equals angle of reflection.
    Hence the reflected ray also makes an angle of 40° with the normal.

WORKSHEET-2

1. **Correct Answer : is (C) 30 cm**

   For a spherical mirror, radius of curvature $R = 2f$.
   So, $R = 2 \times 15 = 30$cm.

2. **Correct Answer : is (C) Radius of curvature**

It is the distance between the pole (P) and the centre of curvature (C).
This distance equals the radius of the sphere forming the mirror.

3. **Correct Answer : is (D) Between F and P**

A concave mirror forms a virtual, erect, and magnified image in this position.
This enlarged image makes shaving easier.

4. **Correct Answer : is (C) Images are always virtual,**

erect, and diminished behind the mirror
Convex mirrors always produce virtual and upright images.
The image size is reduced, giving a wider field of view.

5. **Correct Answer : is (C) Convex mirror**

Convex mirrors provide a wide field of view by diverging light rays.
This helps drivers see more area behind the vehicle.

6. **Correct Answer : is (C) Concave mirror**

A concave mirror gives a magnified image when the object is close.
This helps dentists examine teeth clearly.

7. **Correct Answer : is (C) Reflect light effectively for illumination**

When the light source is at the focus, reflected rays become parallel.
This creates a strong, directed beam used in searchlights.

8. **Correct Answer : is (C) Rear-view mirrors in vehicles**

Convex mirrors are used because they show a larger area.
They form smaller images, improving visibility.

9. **Correct Answer : is (C) Image movement is**

small for large object movement
Images formed are diminished, so distance changes appear smaller.
Hence judging actual distance becomes difficult.

10. **Correct Answer : is (B) Receive and send radio signals**

Parabolic mirrors focus signals at the focus point.
This allows efficient transmission and reception of radio waves.

WORKSHEET-3

1. **Correct Answer : (B)**

Light enters a denser medium (glass), so its speed decreases.
Hence the ray bends towards the normal.

2. **Correct Answer : (C)**

Refractive index $n = \frac{3\times10^8}{2\times10^8} = 1.5$.
It is the ratio of speed in vacuum to speed in the medium.

3. **Correct Answer : (B)**

Absolute refractive index is the ratio of speed of light in vacuum to that in the medium.
It shows how much the medium slows down light.

4. **Correct Answer : (C)**

Lateral shift is the sideways displacement of the ray after emerging from the slab.
It is measured as the perpendicular distance between incident and emergent paths.

5. **Correct Answer : (C)**

Apparent depth = Real depth ÷ Refractive index.
Refraction makes objects appear shallower than their real position.

6. **Correct Answer : (C)**

Sunlight bends continuously due to layers of different atmospheric density.
This causes early sunrise and late sunset.

7. **Correct Answer : (B)**

Apparent depth $= 12/1.5 = 8$cm.
The object appears closer to the surface due to refraction.

8. **Correct Answer : (C)**

At the critical angle, refraction angle becomes 90°.
Beyond this, total internal reflection occurs.

9. **Correct Answer : (A)**

A prism is bounded by three plane surfaces. Two surfaces are refracting faces and one is the base.

10. **Correct Answer : (C)**

Violet light travels slower than red in a medium.
Hence violet has a greater refractive index than red.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer : is C) Because only a portion of the**

incident light is reflected and a large portion enters the water.
Only part of the light reflects from the water surface while the rest refracts into water. Hence the image appears less bright.

2. **Correct Answer : is B) Concave**

A concave mirror forms a magnified image when the object is close to it. Slightly curved glass surfaces can act like concave mirrors.

3. **Correct Answer : is A) Concave mirror**

When a point source is placed at the focus of a concave mirror, reflected rays become parallel. This is used in torches and headlights.

4. **Correct Answer : is A) Under no circumstances**

A convex mirror always forms a virtual, erect, and diminished image. It never produces an inverted image.

5. **Correct Answer : is C) For both convex and concave mirrors**

For all spherical mirrors, the focal length equals half the radius of curvature. This relation comes from mirror geometry.

6. **Correct Answer : is B) Concave mirror**
A concave mirror can produce a virtual, erect, magnified image when the object lies between pole and focus. Other mirrors cannot do this.

7. **Correct Answer : is B) Violet**
Violet light bends the most during refraction. Greater bending indicates a higher refractive index.

8. **Correct Answer : is B) 4 cm towards the mirror**
In a plane mirror, image distance equals object distance. So when the mirror moves 2 cm, the image shifts by twice that distance.

9. **Correct Answer : is A) Aperture**
The aperture is the diameter of the reflecting surface of a spherical mirror. It determines the amount of light reflected.

10. **Correct Answer : is C) Principal axis**
The line joining the pole and centre of curvature is called the principal axis. It is the main reference line in ray diagrams.

11. **Correct Answer : is A) Pole**
The pole is the midpoint of the reflecting surface of a spherical mirror. All mirror measurements are taken from this point.

12. **Correct Answer : is A) 32 cm**

For spherical mirrors, $R = 2f$. With $f =$ 16cm, the radius of curvature is 32 cm.

13. **Correct Answer : B) Convex**

A convex mirror has its reflecting surface curved outward, i.e., on the convex side.
Because of this outward curvature, it always forms a virtual, erect, and diminished image.

14. **Correct Answer : C) 40 cm**

For a spherical mirror, the radius of curvature $R$is twice the focal length f. Given $PF = f = 20$ cm, therefore $R = 2f = 40$ cm.

15. **Correct Answer : B) Real**

A concave mirror converges parallel rays to a point in front of the mirror. Therefore its principal focus is real.

16. **Correct Answer : B) Between principal focus and pole**

A convex mirror always forms a virtual, erect, and diminished image. The image lies between the pole and focus.

17. **Correct Answer : A) Extremely diminished**

For an object at infinity, rays are parallel and appear to meet at the focus. Hence the image is highly diminished.

18. **Correct Answer : B) Virtual, magnified**

When the object is placed within the focal length of a concave mirror, the image formed is virtual and enlarged.

19. **Correct Answer : C) Concave mirror**

A concave mirror concentrates sunlight at its focus. This produces high temperature required in a solar furnace.

20. **Correct Answer : C) Angle of refraction**

The angle formed between the refracted ray and the normal is called the angle of refraction.

21. **Correct Answer : B) sin i / sin r = constant**

According to Snell's law, the ratio of sine of angle of incidence to sine of angle of refraction remains constant for two media.

22. **Correct Answer : A) Greater**

Light travels slower in a denser medium. Therefore its refractive index is greater than that of a rarer medium.

23. **Correct Answer : A) VIBGYOR**

The colours in the visible spectrum appear in the order Violet, Indigo, Blue, Green, Yellow, Orange, Red.

24. **Correct Answer : A) 3×10^8 m/s**

The speed of light in vacuum or air is maximum and equals approximately 3×10^8m/s.

25. **Correct Answer : B) Violet**

Violet light has the shortest wavelength and bends the most in a medium, so its speed is minimum.

ONE OR MORE THAN ONE CORRECT OPTION

26. **Correct Answer: A) Plane mirror, B) Convex mirror**

Explanation: Both plane and convex mirrors always form virtual and erect images for any object position. A concave mirror does not always form such images.

27. **Correct Answer: A) It can form a real image, B)**

It can form a virtual image
Explanation: A concave mirror forms real and inverted images when the object is beyond the focus, and virtual and erect images when the object is between the focus and the mirror.

28. **Correct Answer: A) Plane mirror, C) Convex mirror**

Explanation: Plane and convex mirrors always produce erect images irrespective of distance. A concave mirror can form inverted images, so it is not correct here.

29. **Correct Answer: A), B), D)**

Explanation: Light travels slower in water than in air, and its speed is maximum in vacuum. The angle between the incident ray and the normal is called the angle of incidence; light bends away from the normal when going from glass to air.

30. **Correct Answer: B) Glass is optically denser than water, C) Water is optically rarer than glass**

Explanation: Optical density is higher where light travels slower. Since light speed is less in glass than in water, glass is optically denser and water is optically rarer.

31. Correct Answer: A) Refraction of light

Explanation: Atmospheric refraction bends sunlight toward Earth, making the sun appear above the horizon earlier at sunrise and later at sunset.

ASSERTION AND REASON TYPE

32. Correct Answer: A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

A rectangular glass slab has two boundaries — air → glass and glass → air.
The light ray bends at both interfaces due to change in speed, so refraction occurs twice. Hence the reason correctly explains the assertion.

33. Correct Answer: A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

Refractive index $n = \frac{c}{v}$, where $c$is speed of light in vacuum.
Since light travels fastest in vacuum, $v \leq c$, so the refractive index is always greater than or equal to 1.

34. Answer: A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

The apparent size of an object depends on the angle it subtends at the eye.
As distance increases, the viewing angle decreases, so the image appears smaller.
Thus the reason explains the assertion.

35. Correct Answer: A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

For any spherical mirror, the principal focus lies midway between the pole and center of curvature.
Therefore, the focal length is half the radius of curvature, expressed as $f = R/2$.

MATRIX MATCHING TYPE

36. Correct Answer: a–R, b–P, c–S, d–Q

37. Correct Answer: a–Q, b–R, c–P, d–S

COMPREHENSION TYPE

38. Correct Answer: B) Bends towards the normal

Explanation:
When light travels from a rarer medium (air) to a denser medium (like water or glass), it slows down and bends towards the normal.

39. Correct Answer: A) 0°

Explanation:
At normal incidence (0°), the light ray travels straight without bending, so it goes undeviated.

40. Correct Answer: A) Bends away from the normal

Explanation:
When light goes from a denser medium to a rarer medium, it speeds up and bends away from the normal.

41. Correct Answer is C)

Refractive index of glass $\mu = 1.5$

Using formula:

$$v = \frac{c}{\mu} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ m/s}$$

42. Correct Option: A) $^1\mu_2 = \frac{v_1}{v_2}$

Refractive index of medium 2 w.r.t medium 1 =

$$^1\mu_2 = \frac{v_1}{v_2}$$

43. Correct Answer: B) Rarer

- Light travels faster in a rarer medium
- Light travels slower in a denser medium

INTEGER TYPE

44. Correct Answer: Refractive index is the ratio of speed of light in vacuum to that in the medium

$$n = \frac{3\times10^8}{225\times10^8} = \frac{3}{2.25} = 1.33$$

45. Correct Answer: Two inclined surfaces

Explanation: A prism has two slant (inclined) faces that form the refracting angle.

46. Correct Answer: 1

Explanation: When light does not bend, both angles are equal, so refractive index becomes 1.

Using Snell's Law:

$$n = \frac{\sin i}{\sin r}$$

If $i = r$, then

$$n = \frac{\sin i}{\sin i} = 1$$

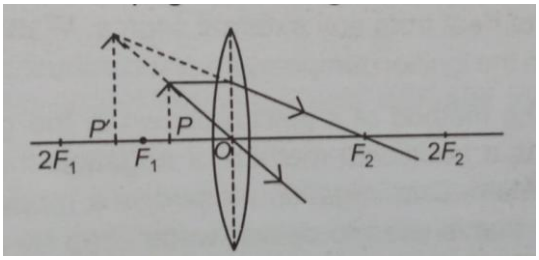
PREVIOUS YEAR QUESTIONS

1. Correct Answer: (A) :

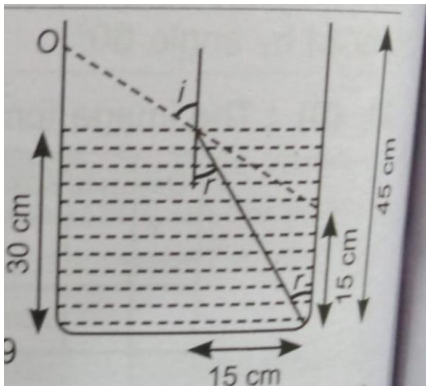
When the mirror is turned by an angle θ, the reflected ray is rotated by an angle 2θ. Hence the reflected ray will be rotated by angle 60°.

2. Correct Answer: (A)

Converging lens is a convex lens. When the object is placed in between focus and pole of the lens, the image obtained is virtual, upright and magnified.



3. Correct Answer: (A) : From Snell's law,



$$1\sin\, i = \mu \sin\, r$$

$$\sin\, 45° = \mu \times \frac{15}{\sqrt{(15)^2 + (30)^2}} \Rightarrow \mu = 1.58$$

Given that, $\mu = \frac{x}{50} \Rightarrow x = 1.58 \times 50 = 79$

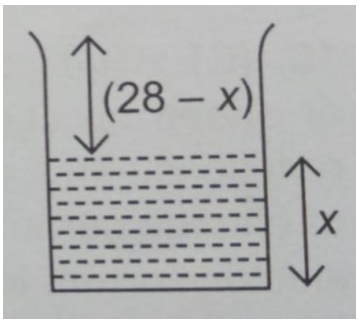
4. Correct Answer: is (B) :

The rainbow colours obtain in order VIBGYOR (from lower to higher). Only in prism (ii) inverted rainbow will obtained with third colour as blue, which is a color of sky.

5. Correct Answer:

Let container is filled up to height **x**.

Thus for student,

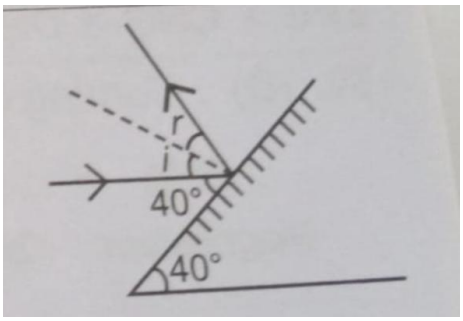


$$\frac{x}{\mu} = 28 - x$$

$$\frac{3}{4}x + x = 28$$

$$\Rightarrow x = 16 \text{ m}$$

6. Correct Answer: (C) : Initially,



$$i = 90° - 40° = 50°$$

Now inclination is 52°,
so new angle of incidence is

$$i = 90° - 52° = 38°$$

So, change in angle of incidence

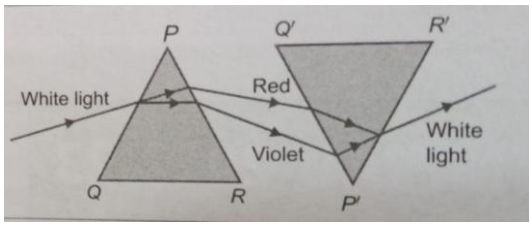
$$= 50° - 38° = 12°$$

Answer: 12°

7. Correct Answer: (A) :

When the source of light moves away from the lens, the intensity remains same as source is at infinity.

8. Correct Answer: (C):

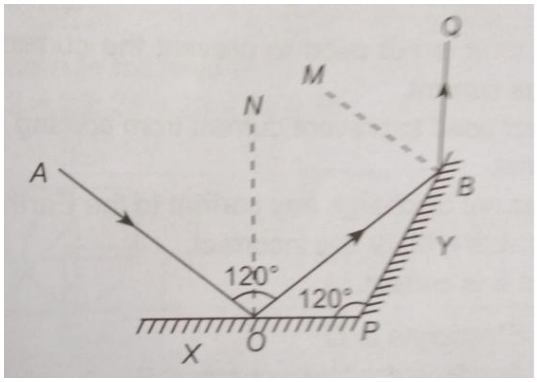


It is the recombination of white light and it can be done by using an inverted prism as shown in the figure.

9. Given,

$$\angle AOB = 120^\circ$$

From Snell's law,
Angle of incidence = Angle of reflection



So,

$$\angle AON = \angle NOB = \frac{120^\circ}{2} = 60^\circ$$
$$\angle BOP = 90^\circ - 60^\circ = 30^\circ$$
$$\angle PBO = 180^\circ - 120^\circ - 30^\circ = 30^\circ$$

So,

$$\angle OBM = \angle MBQ = 90^\circ - 30^\circ = 60^\circ$$

Hence,

$$\boxed{\angle OBM = \angle MBQ = 60^\circ}$$

5. Sound

Solutions

WORKSHEET – 1

1. **Correct Answer: (A) Sound needs a medium to propagate**

   Explanation: Sound travels through a material medium like air. When air is removed, sound cannot travel, so it becomes inaudible.

2. **Correct Answer: (B) 5 s**

   Explanation: Time = Distance ÷ Speed = 1700 ÷ 340 = 5 s. Therefore, the sound reaches after 5 seconds.

3. **Correct Answer: (C) Iron**

   Explanation: Sound travels fastest in solids due to closely packed particles. Hence, it reaches first through iron.

4. **Correct Answer: (A) 8 compressions and 8 rarefactions**

   Explanation: One vibration produces one compression and one rarefaction. So, 8 vibrations produce 8 of each.

5. **Correct Answer: (B) P has lower pitch than Q; R has similar pitch as Q**
   Explanation: More water lowers the frequency and pitch. Equal water levels produce similar pitch.

6. **Correct Answer: (C) 24**
   Explanation: Each wave has one compression and one rarefaction. Thus, 12 waves form 24 regions in total.

7. **Correct Answer: (C) 1200**
   Explanation: Total vibrations = frequency × time = 150 × 8 = 1200. So, 1200 vibrations occur.

8. **Correct Answer: (A) Tightening changes vibration behavior of vocal cords**

   Explanation: Tightening alters how vocal cords vibrate. This changes the quality of the voice produced.

9. **Correct Answer: (C) Sound will not be produced**

   Explanation: Air must pass through the slit to vibrate vocal cords. If blocked, sound cannot be produced.

10. **Correct Answer: (B) Windpipe**

    Explanation: The voice box is located at the upper end of the windpipe. It contains the vocal cords responsible for sound production.

WORKSHEET-2

1. **Correct Answer: (C) 4 Hz**

   Frequency is the reciprocal of time period, $f = 1/T$.
   So, $f = 1/0.25 = 4$Hz.

2. **Correct Answer: Answer: (D) Wavelength**

   The distance between two consecutive crests represents one complete wave.
   This distance is called wavelength.

3. **Correct Answer: Answer: (B) Sound B is**

   louder because it has greater amplitude
   Loudness depends on amplitude of vibration.
   Greater amplitude produces louder sound.

4. **Correct Answer: Answer: (B) Its larger vibrating surface moves more air**

   A larger vibrating surface creates stronger air vibrations.
   This results in greater loudness.

5. **Answer: (C) Pitch (shrillness)**

   Pitch depends on frequency, not amplitude.
   Different frequencies produce different pitch.

6. **Correct Answer: (C) 4 Hz**

Frequency = number of vibrations per second.
$120 \div 30 = 4$Hz.

7. **Correct Answer: (C) 4 times**

Loudness is proportional to the square of amplitude.
Doubling amplitude increases loudness by $2^2 = 4$times.

8. **Correct Answer: (C) 100 times**

An increase of 10 dB means 10 times increase in intensity.
Therefore, 20 dB corresponds to 100 times increase.

9. **Correct Answer: (B) Frequency decreases and pitch becomes low**

Closing more holes increases the vibrating air column length.
Longer air columns produce lower frequency and pitch.

10. **Correct Answer: (C) Difference in quality (timbre)**

Quality (timbre) depends on waveform and overtones.
It allows sounds with same pitch and loudness to be distinguished.

WORKSHEET-3

1. **Correct Answer: (C) 3**

The middle ear has three bones that pass vibrations one after another.
So, the vibration is transmitted in 3 steps before reaching the oval window.

2. **Correct Answer: (C) Pinna → Eardrum → Middle ear bones → Cochlea → Auditory nerve →Brain**

This is the correct pathway of sound from collection to interpretation.
Each part transfers or converts sound until the brain interprets it.

3. **Correct Answer: (C) Conversion of vibrations into electrical signals**

Cochlear fluid movement stimulates sensory cells to create nerve signals.
If fluid movement stops, electrical signal formation fails.

4. **Correct Answer: (B) Bone conduction through skull vibrations**

Vibrations travel through solid objects and skull bones to the inner ear.
This allows hearing even when air transmission is reduced.

5. **Correct Answer: (C) 60 dB**

Painful sound level = 120 dB and conversation = 60 dB.
Difference = 120 – 60 = 60 dB.

6. **Correct Answer: (B) Excessive or** unwanted sounds in the environment
Noise pollution refers to disturbing or harmful sounds.
It can affect health and hearing.

7. **Correct Answer: (B) Loudness, not the** type of sound, can cause harm
Any sound can be harmful if its loudness is very high.
Long exposure to loud music can irritate or damage ears.

8. **Correct Answer: (B) Planting trees along roads and boundaries**

Trees absorb and block sound waves.
This reduces noise reaching nearby homes.

9. **Correct Answer: (B) 20**

Each vibration is amplified once by the middle ear.
So, 20 vibrations produce 20 amplified vibrations.

10. **Correct Answer: (C) Electrical signals will not reach the brain**

The auditory nerve carries signals from the ear to the brain.
If it fails, sound cannot be interpreted even though other parts work.

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: B) BD**

Explanation:
Half wavelength is the distance between a crest and the next trough. In the diagram, B marks the crest and D marks the trough, so BD represents half the wavelength.

2. Correct Answer: D) D

Explanation:
A trough is the lowest point of a wave below the equilibrium line. In the diagram, point D is at the lowest position, so it represents the trough.

3. Correct Answer: B) 1.5 m

Frequency = 320 ÷ 2 = 160 Hz.
Wavelength = velocity ÷ frequency = 240 ÷ 160 = 1.5 m.

4. Correct Answer: A) compression

In a longitudinal wave, particles are closest at compressions.
Hence density is maximum at compressions.

5. Correct Answer: C) 60 vibrations per minute

1 hertz means 1 vibration per second.
In one minute (60 s), it equals 60 vibrations.

6. Correct Answer: C) Hertz

Frequency measures number of vibrations per second.
Its SI unit is Hertz (Hz).

7. Correct Answer: C) Amplitude

The greatest displacement on either side of mean position is amplitude.
It represents the energy of the wave.

8. Correct Answer: A) solids

Sound travels fastest in solids because particles are tightly packed.
Closer particles transfer vibrations more quickly.

9. Correct Answer: A) 0 to 80 dB

Sounds up to about 80 dB are generally considered safe for hearing.
Higher levels may damage the ear with long exposure.

10. Correct Answer: D) decibel

Loudness is measured in decibel (dB).
It indicates the intensity level of sound.

11. Correct Answer: D) Tuning fork

A tuning fork produces sound of a single frequency.
Hence it gives a pure tone.

12. Correct Answer: A) 1:1

Pitch depends on frequency, not amplitude.
Since pitch is same, their frequencies are equal.

13. Correct Answer: A) Higher than 20,000 Hz

Ultrasound refers to sound frequencies above human hearing range.
So it must be greater than 20 kHz.

14. Correct Answer: D) all of these

Noise pollution can cause lack of sleep, anxiety, and hypertension.
Continuous loud sound affects mental and physical health.

15. Correct Answer: C) noise pollution

Excess unwanted sound in surroundings is called noise pollution.
It disturbs normal living conditions.

16. Correct Answer: C) Trumpet

Higher frequency means higher pitch.
500 Hz (trumpet) is highest among the given frequencies.

17. Correct Answer: C) A man

Lower frequency corresponds to deeper voice.
Adult men usually have the lowest voice frequency.

18. Correct Answer: A) 2.02×10^4 hertz

Frequency $f = \frac{v}{\lambda} = \frac{343.4}{1.7\times10^{-2}} = 2.02 \times 10^4$ Hz.
Frequency is inversely proportional to wavelength when velocity is constant.

19. Correct Answer: B) 1320 m

Distance = speed × time = $330 \times 4 = 1320$m.
Light reaches almost instantly, so only sound travel time is considered.

20. Correct Answer: B) the number of

complete vibrations per second
Frequency tells how many oscillations occur in one second.
Its SI unit is hertz (Hz).

21. Correct Answer: B) longitudinal

In air, sound travels through compressions and rarefactions of particles.
Hence the wave is longitudinal, not transverse.

22. Correct Answer: B) Parallel

In longitudinal waves, particles vibrate back and forth along the wave direction.
The vibration and propagation directions are the same.

23. Correct Answer: B) Shrill

Plucking closer to the fixed end increases frequency of vibration.
Higher frequency produces a shrill (high-pitched) sound.

24. Correct Answer: B) Jar B

Explanation:
When air is blown across the mouth, the pitch depends on the length of the air column. Jar B has less air space (more water), so the air column is shorter and produces a higher-pitched sound.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

25. Correct Answer: C

Explanation: Wavelength is the distance between two successive similar points in a wave, such as compression to compression or crest to crest. It represents one complete wave cycle in space.

26. Correct Answer: B, C

Explanation: 1 hertz means 1 vibration per second. This equals 60 vibrations per minute and also 120 vibrations in 2 minutes.

27. Correct Answer: B, C

Explanation: Frequency is measured as the number of vibrations per second, so its unit is 1/second (s^{-1}), called Hertz (Hz).
Centimetre measures length and decibel measures loudness.

28. Correct Answer: A

Explanation: Quality (timbre) of sound depends on the mixture of different frequencies produced by a source. This allows us to distinguish sounds even when pitch and loudness are the same.

ASSERTION AND REASON TYPE

29. Correct Answer: (C) A is true, but R is false

The outer ear collects sound waves and directs them toward the eardrum, so the assertion is correct.
However, the outer ear is funnel-shaped (not ring-shaped), which helps in gathering sound.

30. Correct Answer: (D) A is false, but R is true

The cochlea converts sound vibrations into electrical nerve signals, not electromagnetic signals.
It contains fluid and hair cells that bend with fluid movement, which is correctly stated in the reason.

31. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A

Pitch depends on the frequency of vibration; higher frequency means higher pitch.
A higher frequency produces a shrill (high-pitched) sound, correctly explaining the assertion.

32. Correct Answer: (A) Both A and R are true, and R is the correct explanation of A

Even with the same pitch and loudness, sounds can differ due to quality (timbre).
Different overtones and their amplitudes give each sound source a unique character.

MATRIX MATCHING TYPE

- Pitch → Q) Frequency
  Explanation: Pitch depends on the frequency of vibration. Higher frequency means higher pitch.
- B) Unit of Frequency → P) Hertz
  Explanation: Frequency is measured in hertz (Hz).
- C) Loudness → S) Amplitude
  Explanation: Loudness depends on the amplitude of vibration.
- D) Quality → R) Wave form
  Explanation: Quality (timbre) depends on the waveform of the sound.

33. A–Q, B–P, C–S, D–R

- A. Source of every sound → Q. Vibrating body
  Explanation: Sound is produced by vibrating bodies.
- B. Sound in vacuum → P. Sound cannot travel
  Explanation: Sound needs a material medium to travel.
- C. More water in jalatarang bowl → R. Lower pitch
  Explanation: More water lowers the frequency, producing a lower pitch.
- D. Sound travels fastest → S. In solids
  Explanation: Sound travels fastest in solids because particles are closely packed.

Answer:

34. Correct Answer: A–Q, B–P, C–R, D–S

COMPREHENSION TYPE

35. Correct Answer: B) Amplitude

Loudness depends mainly on the amplitude of vibrations. Greater amplitude means more energy is carried by the sound wave, making it louder.

36. Correct Answer: C) Shriller

Pitch depends on frequency, and higher frequency produces a higher pitch. High-pitched sounds are heard as shrill sounds.

37. Correct Answer: C) 20 Hz – 20,000 Hz

The human ear can detect sounds only within this frequency range. Sounds below 20 Hz are infrasonic and above 20,000 Hz are ultrasonic, so they are not audible to humans.

38. Answer: C) Different Loudness and Same Pitch

Sounds A and B have the same frequency (same number of waves), so their pitch is the same.
B has greater amplitude than A, so it is louder.

39. Answer: C) Different Loudness and Different Pitch

B has larger amplitude than C, so their loudness is different.
C has higher frequency (more cycles), so its pitch is higher than B.

40. Answer: A) same loudness but different pitch

A and C have nearly the same amplitude, so their loudness is the same.
C has higher frequency than A, therefore it has a higher pitch.

INTEGER TYPE

41. The speed of sound in air is nearly *340 m/s*.

Explanation: Sound travels in air at about 340 m/s at room temperature. This value may change slightly with temperature and humidity.

42. If the amplitude of a sound wave becomes 3 times, the loudness increases by *9 times*.

Explanation: Loudness is proportional to the square of amplitude. So, when amplitude becomes 3 times, loudness becomes $3^2 = 9$times.

43. Sound B is *16 times* louder than Sound A.
Explanation: Loudness ∝ (amplitude)², so ratio = $4^2 : 1^2 = 16 : 1$. Therefore, sound B is 16 times louder than sound A.

PREVIOUS YEAR QUESTIONS

1. Correct Answer: (A) : Wavelength is the distance travelled by one wave.

So, it is λ = 2.7 − 1 = 1.7 cm.

2. Correct Answer: (C) : As v ∝ 1/λ,

if frequency is doubled, then wavelength is halved i.e., λ′ = 1.7/2 = 0.85 cm

3. Correct Answer: (D) :

From the given displacement versus distance graph of sound wave, the wave length of sound is 0.04 m.
Speed of wave, v = 320 m/s
As, v = frequency × wavelength

∴ Frequency = 320 / 0.04 = 8000 Hz

4. Correct Answer: (D) : Given, frequency range is 20 Hz to 20 KHz.
Speed of sound is 340 m/s

$\text{Shortest wavelength} = \frac{v}{\nu} = \frac{340}{20\times10^3} = 0.017\,m$

$\text{Longest wavelength} = \frac{340}{20} = 17\,m$

Pitch of the sound increases with increase in frequency.
Hence, (i) – T, (ii) – T, (iii) – T, (iv) – F

5. Correct Answer: (C) : In all the three graphs I, II and IV the waves are following a

particular pattern. In the graph III, no pattern is followed, thus it represents a noise.

6. Correct Answer: (A) :

The graph having longest time period, will have the shortest frequency and hence the lowest pitch. It is clear that, the graph II is having lowest pitch.

7. Correct Answer: (B) : Distance = 5 km = 5 × 1000 m = 5000 m
Time = 100 s

Speed of sound = Distance / time = 5000 / 100 = 50 m s^{-1}

8. Correct Answer: (D) :

Waves transport energy from one place to another.

9. Correct Answer: is A)

6. Electricity and Chemical Effects of Electric Current

Solutions

Worksheet – 1

1. **Correct Answer : C) Silk cloth gains electrons from glass rod**

   Explanation: During rubbing, electrons transfer from glass to silk → glass becomes $+q$, silk becomes $-q$based on conservation of charge.

2. **Correct Answer : B) It loses electrons**

   Explanation: Positive charge means deficiency of electrons → $q = +ne$when electrons are removed.

3. **Correct Answer : C) Conservation of charge**

   Explanation: Total charge before rubbing = total charge after rubbing → $\sum q_{initial} = \sum q_{final}$.

4. **Correct Answer : C) Charge resides only on the outer surface**

   Explanation: In electrostatic equilibrium, free charges move to surface due to mutual repulsion → $E_{inside} = 0$.

5. **Correct Answer : C) Move towards the rod**

   Explanation: Due to electrostatic induction, opposite charges attract → $F = k\frac{q_1 q_2}{r^2}$.

6. **Correct Answer : B) Charging by conduction**

   Explanation: Direct contact allows transfer of electrons between bodies.

7. **Correct Answer : C) Induction**

   Explanation: Induction involves redistribution of charges without contact.

8. **Correct Answer : B) Presence of electric charge**

   Explanation: Like charges on leaves repel → $F \propto \frac{1}{r^2}$causing divergence.

9. **Correct Answer : B) Converge**

   Explanation: Opposite charges neutralize → net charge decreases → leaf divergence reduces.

10. **Correct Answer : C) Clouds contain electric charge**

    Explanation: Spark discharge occurs when potential difference exceeds breakdown value → $V = IR$conceptually indicates stored electric charge.

WORKSHEET-2

1. **Correct Answer : B) 2 A**

   Explanation: $I = \frac{Q}{t}$

   $$I = \frac{6\ \text{C}}{3\ \text{s}} = 2\ \text{A}$$

2. **Correct Answer : B) $1\ \text{C s}^{-1}$**

   Explanation: $I = \frac{Q}{t}$

   $$1\ \text{A} = 1\text{C s}^{-1}$$

3. **Correct Answer : B) 5×10^{-3} A**

   Explanation: $1\ \text{mA} = 10^{-3}\ \text{A}$

   $$5\ \text{mA} = 5 \times 10^{-3}\ \text{A}$$

4. **Correct Answer : B) Positive terminal to negative terminal**

   Explanation: Conventional current flows from higher potential to lower potential

5. **Correct Answer : B) Potential difference becomes zero**

   Explanation: $I = \frac{V}{R}$ If $V = 0$, then $I = 0$

6. **Correct Answer : C) Electric cell**

   Explanation: A cell maintains potential difference

   $$V = \frac{W}{Q}$$

7. **Correct Answer : C) Electromotive force**
Explanation: EMF $= \frac{W}{Q}$
8. **Correct Answer : B) Copper to zinc**
Explanation: Current flows in external circuit from positive to negative terminal
9. **Correct Answer : C) Zinc container**
Explanation: In a dry cell, zinc acts as the negative terminal (anode)
10. **Correct Answer : B) Increase voltage**
Explanation: $V_{\text{total}} = V_1 + V_2 + V_3 + \cdots$

WORKSHEET-3

1. **Correct Answer : B) Higher resistance**

Explanation: The heat produced in a conductor when electric current passes through it is given by Joule's law of heating: $H = I^2Rt$
where $H$is heat produced, $I$is current, $R$is resistance, and $t$is time.
For the same current and time, heat produced is directly proportional to resistance.
Since nichrome has higher resistance than copper, it produces more heat and becomes hotter.

2. **Correct Answer : C) Heating effect of current**

Explanation: When electric current flows through a conductor, electrical energy is converted into heat energy due to resistance of the conductor.
The amount of heat produced is given by:
$$H = I^2Rt$$
This heating effect causes plastic parts in plugs and sockets to melt during excessive current flow.

3. **Correct Answer : C) Current produces magnetic field**

Explanation: When electric current flows through a conductor, a magnetic field is produced around it.
The strength of the magnetic field depends on the current:
$$B \propto I$$
This magnetic field causes the compass needle to deflect when placed near a current-carrying wire.

4. **Correct Answer : C) Electromagnet**
Explanation: When electric current passes through a coil of insulated copper wire wound around an iron rod, a magnetic field is produced which magnetizes the iron rod.
5. **Correct Answer : C) Increasing number of turns in coil**

Explanation: The strength of an electromagnet depends on the number of turns in the coil and the current flowing through it:
$$B \propto nI$$

6. **Correct Answer : C) Is weakly ionized**

Explanation: Electric current in liquids flows due to movement of ions. Distilled water has very few ions because it is weakly ionized, so it does not conduct electricity effectively.

7. **Correct Answer : B) Salt provides ions for conduction**
Explanation: When salt is dissolved in distilled water, it dissociates into positive and negative ions.
8. **Correct Answer : C) Electrolysis**
Explanation: Electrolysis is the chemical decomposition of a compound in molten or dissolved state by passing electric current through it.
9. **Correct Answer : B) Cathode**
Explanation: During electrolysis, positively charged hydrogen ions H^+move towards the negatively charged electrode (cathode).
10. **Correct Answer : C) Positive terminal**
Explanation: In electroplating, the metal to be deposited (copper) is connected to the positive terminal (anode).

COMPETITIVE CORNER

1. **Correct Answer : B) Lost electrons**

Explanation: A positively charged rod attracts electrons from the metal sphere on contact. Loss of electrons leaves the sphere positively charged, causing electroscope leaves to diverge.

2. Correct Answer : B) Potential difference became equal

Explanation: When the battery's chemical energy is used up, it cannot maintain potential difference (V). Since current $I = \frac{V}{R}$, current becomes zero when $V = 0$.

3. Correct Answer : C) Connect more lemon cells in series

Explanation: Connecting cells in series increases total voltage, improving chances of LED glowing.

4. Correct Answer : C) Chemical composition changes permanently

Explanation: Continuous use exhausts reactants in the cell, causing irreversible chemical changes that stop current flow.

5. Correct Answer : B) Exchange positions

Explanation: Reversing battery reverses electrode polarity, so hydrogen and oxygen gases swap electrodes.

6. Correct Answer : B) Magnetic effect depends on current

Explanation: Electromagnet works only when electric current flows through the coil, producing magnetic field.

7. Correct Answer : B) Copper dissolves into solution

Explanation: In electroplating, copper from anode dissolves into electrolyte to maintain Cu^{2+} ion concentration.

8. Correct Answer : C) Current will not pass easily

Explanation: Pure water lacks ions; adding acid increases ionisation and electrical conductivity.

9. Correct Answer : C) Cathode thicker

Explanation: During purification, pure metal deposits on cathode making it thicker over time.

10. Correct Answer : B) Conversion of chemical to electrical energy

Explanation: Electric cell converts chemical energy into electrical energy, enabling LED to glow.

11. Correct Answer : B) Is weakly ionised

Explanation: Distilled water has very few ions, so electrical conductivity is extremely low.

12. Correct Answer : C) Protect from corrosion

Explanation: Electroplating forms protective metal coating preventing rusting.

13. Correct Answer : B) Settle separately

Explanation: Impurities fall as anode mud during electrolytic purification.

14. Correct Answer : C) Increases

Explanation: Heating effect depends on $H = I^2Rt$; increase in current raises temperature.

15. Correct Answer : C) Conversion of mechanical energy

Explanation: Generator converts mechanical energy into electrical energy by electromagnetic induction.

16. Correct Answer : B) Negative terminal

Explanation: Hydrogen gas forms at cathode (negative electrode) during electrolysis of water.

17. Correct Answer : B) Deposits thin metal layer

Explanation: Electroplating coats object with thin shiny metal layer improving appearance.

18. Correct Answer : B) 3 A

Explanation: $I = \frac{Q}{t} = \frac{12}{4} = 3\ A$

19. A) 0.01 C

Explanation: $Q = I \times t = 0.002 \times 5 = 0.01\ C$

20. Correct Answer : A) 0.25 A

Explanation: $Q = 500\ mC = 0.5\ C$

$$I = \frac{Q}{t} = \frac{0.5}{2} = 0.25\ A$$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

21. **Correct Answer : A) Prevent rusting of metals, B) Repairing worn out parts of machinery, C) Ornamentation and decoration purposes**

Explanation: Electroplating is used to coat metals with another metal layer to prevent corrosion, rebuild worn parts, and improve appearance. In food cans, tin is electroplated on iron to prevent rusting of the container, not to preserve food directly. Food preservation occurs due to airtight sealing, not because of electroplating itself.

22. **Correct Answer : B) Hydrogen and C) Oxygen**

Explanation: Electrolysis of water decomposes it into hydrogen gas at cathode and oxygen gas at anode in the ratio 2:1.

23. **Correct Answer : B) Distilled water + salt and D) Distilled water + Lemon juice**

Explanation: Adding salt or lemon juice provides ions which increase conductivity of distilled water.

24. **Correct Answer : A) Charge given to an insulator remains stationary at the supplied position, C) Charge given to a conductor resides at the outer surface**

Explanation: In insulators, charges cannot move freely; in conductors, free charges redistribute and reside on the outer surface.

ASSERTION AND REASON TYPE

25. **Correct Answer : A) Both A and R are true and R is the correct explanation of A**

Explanation: When a positively charged rod touches a metal sphere, electrons move from the sphere to the rod due to electron deficiency in the rod. Loss of electrons leaves the sphere positively charged.

26. **Correct Answer : D) A is false but R is true**

Explanation: A dry cell cannot be reused once its chemicals are exhausted because the internal chemical reactions are irreversible.

27. **Correct Answer : C) A is true but R is false**

Explanation: Adding acid increases ionisation and provides more charged particles (ions) to carry current; it does not decrease them.

28. **Correct Answer : A) Both A and R are true and R is the correct explanation of A**

Explanation: Electromagnetism exists only when current flows. Switching OFF current removes magnetism, allowing easy release of scrap.

MATRIX MATCHING TYPE

29. **Correct Answer :**

a – p
b – q
c – r
d – s

30. **Correct Answer :**

a – r
b – q
c – p
d – s

Explanation: During electrolysis, reduction occurs at the cathode causing metal ions to gain electrons and deposit as metal, so it gains metal. In electrolytic purification, the anode undergoes oxidation and dissolves into the solution, becoming thinner. Nichrome wire has high resistance and produces more heat as per heating effect $H = I^2Rt$. A generator converts mechanical energy into electrical energy by electromagnetic induction.

COMPREHENSION TYPE

31. **Correct Answer : B) Rubber**

Explanation: Electric insulators are substances that do not allow electric current to pass through them because they lack free electrons or mobile charge carriers. Rubber is an insulator, so it prevents the flow of electric

current, whereas copper, iron, and metallic bars are conductors.

32. Correct Answer : C) Conductors

Explanation: Substances that allow electric current to pass through them easily due to the presence of free charge carriers (free electrons) are called conductors. Metals are examples of good conductors.

33. Correct Answer : A) Free electrons

Explanation: Metals have loosely bound outer electrons that can move freely within the material. These free electrons drift under the influence of an electric field, enabling the conduction of electric current.

34. Correct Answer : C) To protect the objects from rusting

Explanation: Chromium electroplating deposits a thin protective chromium layer over iron, which prevents contact with air and moisture, thereby avoiding oxidation (rusting) of iron.

35. Correct Answer : A) Electroplating

Explanation: The process of depositing a thin layer of one metal over another using electric current is called electroplating. It improves appearance and protects the base metal from corrosion.

36. Correct Answer : A) Chemical effect of current

Explanation: Electroplating occurs due to chemical reactions produced by the passage of electric current through an electrolyte, which leads to deposition of metal ions on the electrode.

INTEGER TYPE

37. Correct Answer : 2 C

Explanation: Given current $I = 250\ mA = 0.25\ A$, time $t = 8\ s$

Using formula $Q = I \times t$

$$Q = 0.25 \times 8 = 2\ C$$

38. Correct Answer : 2 A

Explanation: Given charge $Q = 6\ C$, time $t = 3\ s$

Using formula $I = \frac{Q}{t}$

$$I = \frac{6}{3} = 2\ A$$

39. Correct Answer : 20 C

Explanation: Given current $I = 5\ A$, time $t = 4\ s$

Using formula $Q = I \times t$

$$Q = 5 \times 4 = 20\ C$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer : C) I, II and III only

Explanation: In the given circuit, the bulb will glow only if the solution allows electric current to pass through it. Substances like sodium chloride (NaCl), copper sulphate ($CuSO_4$), and silver nitrate ($AgNO_3$) dissociate into positive and negative ions when dissolved in water. These free ions act as charge carriers and conduct electricity through the solution, completing the circuit and lighting the bulb. Sugar solution does not ionise in water as it is a covalent compound, so it does not provide free ions for conduction and the bulb will not glow.

2. Correct Answer :A) P-2, Q-3, R-1, S-4

Explanation: In electrolysis, the anode is connected to the positive terminal, while the cathode is connected to the negative terminal of the battery. Ions are atoms or molecules carrying positive or negative charge, and an electrolyte is a conducting solution that allows current to pass through it due to the presence of free ions.

3. Correct Answer :C) (iii) only

Explanation: According to Faraday's law, the mass deposited is **directly proportional** to the quantity of electricity passed ($m \propto Q$), so statement (i) is incorrect. In electroplating, the object to be electroplated is taken as the cathode, not anode, so (ii) is also incorrect. Copper from copper sulphate solution

deposits at the cathode (negative terminal), making (iii) correct.

4. Correct Answer : Answer: B) Q, P and R

Explanation: In the figure, Q represents the solution in which electrodes are dipped, so it is the electrolyte. The electrode connected to the positive terminal of the battery is the anode (P), and the electrode connected to the negative terminal is the cathode (R).

5. Correct Answer : Answer: B) I, III and IV only

Explanation: A fuse is a safety device connected in the live wire of a circuit to protect electrical appliances from damage due to excessive current. It melts and breaks the circuit when current exceeds the safe limit (Statement II is correct). It does not prevent current from coming out of neutral or live terminals (I and III are incorrect), nor does it allow current to discharge to Earth—that is the function of earthing (IV is incorrect).

6. Correct Answer : D) Carbon Anode Cathode Oxygen Hydrogen

Explanation: In electrolysis of acidulated water, inert carbon electrodes are used. The electrode connected to the positive terminal is called the anode, and the one connected to the negative terminal is called the cathode. Oxygen gas is released at the anode due to oxidation, while hydrogen gas is released at the cathode due to reduction.

7. Correct Answer :A) Only Statement I is correct

Explanation: Liquids conduct electricity due to the presence of free ions which carry charge through the solution. The conventional direction of current is the direction of movement of positive charge, which is opposite to the movement of negative ions.

8. Correct Answer : C) To cut off the current when there is a short circuit

Explanation: A circuit breaker is a safety device that automatically breaks the circuit when excess current flows due to short circuit or overload, preventing damage to appliances.

9. Correct Answer : C) Statement 1 is true but statement 2 is false

Explanation: Electric current can cause or speed up chemical reactions, known as the chemical effect of current. Cane sugar solution does not ionise in water and hence does not conduct electricity.

10. Correct Answer : C) Electroplating

Explanation: Electroplating is based on the chemical effect of electric current, not on the magnetic effect of current. Hence, it is incorrectly listed under magnetic effect

7. Stars and Solar System

Solutions

Worksheet – 1

1. **Correct Answer : B) Celestial objects**

   Explanation: The Sun, Moon, stars, and planets are natural objects present in space and are collectively called celestial objects. Satellites are only bodies that revolve around planets; constellations are star patterns; meteoroids are small rocky bodies.

2. **Correct Answer : C) The Sun is at its highest position**

   Explanation: The length of a shadow depends on the Sun's altitude. When the Sun is at its highest position (around noon), the angle of elevation is maximum, so the shadow formed is shortest.

3. **Correct Answer : C) Uttarayanam**

   Explanation: When the apparent position of sunrise shifts gradually towards the north, it is called Uttarayanam (northward movement of the Sun). Dakshinayanam refers to the southward shift.

4. **Correct Answer : B) North–South direction**

   Explanation: For a sundial to work accurately, the base (or gnomon alignment) must be along the North–South direction so that the shadow moves correctly from west to east during the day due to Earth's rotation.

5. **Correct Answer : C) Nuclear fusion of hydrogen**

   Explanation: The Sun produces energy by nuclear fusion, where hydrogen nuclei combine to form helium, releasing enormous energy according to Einstein's equation:
   $E = mc^2$.

6. **Correct Answer : B) Venus – Hottest planet**

   Explanation: Venus is the hottest planet due to its thick carbon dioxide atmosphere causing a strong greenhouse effect. Mercury does not have a thick CO_2 atmosphere; Mars has seasons; Earth has a magnetic field.

7. **Correct Answer : B) Longer by 37 minutes**

   Explanation: Earth's day = 24 hours.
   Mars' day = 24 hours 37 minutes.
   Difference = 37 minutes.
   Therefore, a Martian day is approximately 37 minutes longer than Earth's day.

8. **Correct Answer : C) Have rings and many moons**

   Explanation: The outer planets (Jupiter, Saturn, Uranus, Neptune) are gas giants and generally have ring systems and many natural satellites (moons). They are not rocky and are larger than Earth.

9. **Correct Answer : C) Meteorite**

   Explanation: A meteoroid in space becomes a meteor when it burns in Earth's atmosphere. If it reaches and strikes the Earth's surface, it is called a meteorite.

10. **Correct Answer : D) Cassiopeia**

    Explanation: Cassiopeia is a constellation that appears like a distorted "W" or "M" shape in the night sky. Orion and Ursa Major have different recognizable shapes.

WORKSHEET-2

1. **Correct Answer :**

   Explanation: The Moon's mass ≈ **1/81** of Earth's mass and its diameter ≈ **1/4** of Earth's diameter.

2. **Correct Answer : B) Waning phase**

Explanation: If a half Moon is seen at sunrise overhead, it is the last (third) quarter, which belongs to the waning phase. In this phase, the illuminated part decreases after full Moon.

3. **Correct Answer : C) Full Moon day**

Explanation: On full Moon day, the Moon is opposite the Sun in the sky (elongation ≈ 180°), so it appears farthest from the Sun.

4. **Correct Answer : B) The illuminated half faces away from Earth**

Explanation: On Amavasya (New Moon), the Moon is between Earth and Sun, and the sunlit half faces away from Earth, so it is not visible.

5. **Correct Answer : B) Gibbous**

Explanation: When more than half but not the full illuminated portion is visible, the phase is called gibbous (either waxing or waning gibbous).

6. **Correct Answer : B) Moon comes between Sun and Earth on new Moon day**

Explanation: A solar eclipse occurs only on a new Moon day when the Moon lies between the Sun and Earth and blocks sunlight partially or totally.

7. **Correct Answer : C) Hybrid Solar Eclipse**

Explanation: A hybrid solar eclipse appears total in some regions and annular in others due to variations in Earth–Moon distance and curvature of Earth.

8. **Correct Answer : B) Partial lunar eclipse**

Explanation: If only part of the Moon enters Earth's dark central shadow (umbra), it is called a partial lunar eclipse.

9. **Correct Answer : B) Earth is spherical**

Explanation: Returning to the same place after continuous travel in one direction supports the idea that Earth is spherical in shape.

10. **Correct Answer : C) Solar year – 365¼ days**

Explanation: One solar year (time taken by Earth to complete one revolution around the Sun) is approximately 365.25 days.

WORKSHEET-3

1. **Correct Answer : C) 354 days**

Explanation: One lunar month ≈ 29.5 days.
12 lunar months = 12 × 29.5 = 354 days (approx).

2. **Correct Answer : D) About 33 days**

Explanation: Difference per year = 365 – 354 = 11 days.
In 3 years = 11 × 3 = 33 days.

3. **Correct Answer : B) Because Earth takes about 365¼ days to revolve around the Sun**

Explanation: Earth's revolution period ≈ 365.25 days.
Extra 0.25 day per year accumulates to 1 day in 4 years (0.25 × 4 = 1), so an extra day is added as a leap day.

4. **Correct Answer : B) 1800**

Explanation: In the Gregorian calendar, century years are leap years only if divisible by 400.
1600, 2000, 2400 are divisible by 400 → leap years.
1800 is not divisible by 400 → not a leap year.

5. **Correct Answer : B) Match lunar months with solar seasons**

Explanation: A luni–solar calendar adds an extra month (Adhika Maasa) to adjust the 11-day yearly difference and align lunar months with solar seasons.

6. **Correct Answer : B) Changes slightly but within about a month**

Explanation: In a luni–solar calendar, the festival remains seasonally fixed but its Gregorian date shifts within a range of about one month due to lunar phases.

7. **Correct Answer : C**

8. **Correct Answer : B) Chaitra has 31 days**

   Explanation: In the Indian National Calendar, during a leap year, the month of Chaitra has 31 days instead of 30.

9. **Correct Answer : B) Its Gregorian date may occur in different months over years**

   Explanation: A purely lunar festival shifts across seasons and Gregorian months because the lunar year is shorter by about 11 days each year.

10. **Correct Answer : C) 100 minutes**

    Explanation: Low Earth Orbit satellites (~800 km altitude) complete one revolution in about 90 – 100 minutes due to gravitational force providing centripetal force.

COMPETITIVE CORNER

SINGLE CORRECT OPTION TYPE

1. **Correct Answer : B) It is local noon**

   Explanation: At local noon, the Sun crosses the local meridian, and the shadow of a vertical object falls exactly along the north–south line.

2. **Correct Answer : B) Dakshinayanam**

   Explanation: The apparent southward shift of the sunrise point is called Dakshinayanam. Uttarayanam refers to the northward shift.

3. **Correct Answer : B) Venus**

   Explanation: Venus takes about 243 Earth days to rotate once (length of day) but only about 225 Earth days to revolve around the Sun (length of year). Thus, its day is longer than its year.

4. **Correct Answer : B) Mercury**

   Explanation: Mercury has no significant atmosphere, causing extreme temperature variations between day and night, and it has no natural satellites (moons).

5. **Correct Answer : C) Jupiter**
   Explanation: Jupiter's Great Red Spot is a giant storm that has lasted for more than 300 years.

6. **Correct Answer : C) Has a cloudy atmosphere**
   Explanation: Titan, a moon of Saturn, is unique because it has a thick, cloudy nitrogen-rich atmosphere.

7. **Correct Answer : B) Away from the Sun**
   Explanation: A comet's tail always points away from the Sun due to solar wind and radiation pressure.

8. **Correct Answer : B) Kuiper belt**
   Explanation: The Kuiper Belt lies beyond Pluto's orbit and contains thousands of icy bodies and dwarf planets.

9. **Correct Answer : B) The Earth revolves around the Sun**

   Explanation: The apparent north–south shift of sunrise occurs due to Earth's revolution around the Sun combined with the tilt of Earth's axis.

10. **Correct Answer : D) Ursa Major**

    Explanation: The Pole Star (Polaris) can be located by extending a line through the two pointer stars of Ursa Major.

11. **Correct Answer : B) Sirius**

    Explanation: Sirius is the brightest star in the night sky and is located near the constellation Orion.

12. **Correct Answer : A) Appears closer to the Sun in the sky each morning**

    Explanation: During the waning phase, the Moon moves closer to the Sun in the sky and rises later each night, becoming visible mainly in the early morning.

13. **Correct Answer : C) Sun, Earth, and Moon are not always perfectly aligned**

    Explanation: A lunar eclipse occurs only when the Sun, Earth, and Moon align exactly at full Moon. Usually, the Moon's orbit is slightly tilted, preventing alignment.

14. **Correct Answer : B) 88 days**
Explanation: Difference per year = 365 – 354 = 11 days.
For 8 years = 11 × 8 = 88 days.
15. **Correct Answer : C) Chaitra**
Explanation: In the Indian National Calendar, Chaitra has 31 days in a leap year instead of 30.
16. **Correct Answer : C)**
17. **Correct Answer : C) 100 minutes**
Explanation: Artificial satellites at about 800 km altitude (Low Earth Orbit) typically complete one revolution in about 90–100 minutes.
18. **Correct Answer : B) Moon's reflection of sunlight and its position relative to Earth and Sun**
Explanation: The Moon does not produce light; its changing phases occur due to reflected sunlight and its changing position relative to Earth and the Sun.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

19. **Correct Answer : A, B, D**

Explanation:

A) Mercury has a very thin atmosphere called an exosphere.
B) Mercury has no natural satellites.
C) Incorrect. Mercury's rotational period is about 59 Earth days and its revolution period is about 88 Earth days, so its day is not shorter than its year.
D) Due to the absence of a thick atmosphere, Mercury experiences extreme temperature variations between day and night.

20. **Correct Answer : A, B, C**

Explanation:
A) All outer planets have ring systems.
B) Outer planets rotate rapidly compared to inner planets.
C) They are mainly gas or ice giants composed largely of hydrogen, helium, and other volatiles.
D) Incorrect. Outer planets are much larger than inner planets.

21. **Correct Answer : A, B, D**

Explanation:
A) A lunar year is about 12 × 29.5 days, approximately 354 days.
B) A solar year is about 365 days.
D) The difference 365 – 354 equals about 11 days per year, causing the shift of lunar months with respect to seasons.
C) Earth's rotation does not cause seasonal shifting.

22. **Correct Answer : A, B, D**

Explanation:
A) Comets are composed mainly of rock, dust, and ice.
B) The comet's tail always points away from the Sun due to solar wind and radiation pressure.
C) Incorrect. Comets mainly originate from the Kuiper Belt and Oort Cloud, not only the asteroid belt.
D) When near the Sun, ice sublimates and forms gas and dust tails.

ASSERTION AND REASON TYPE

23. **Correct Answer : A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**

Explanation: At local noon, the Sun attains its maximum altitude in the sky. When the Sun is highest, the angle of elevation is maximum, resulting in the shortest possible shadow of a vertical object.

24. **Correct Answer : D) Assertion is false, but Reason is true.**

Explanation: A solar eclipse occurs only on a new Moon day, not on a full Moon day. The reason statement is correct because during a new Moon, the Moon comes between the Sun and the Earth, which is the required condition for a solar eclipse.

25. Correct Answer : C) Assertion is true, but Reason is false.

Explanation: Lunar m

onths shift with respect to seasons because a lunar year is about 354 days and a solar year is about 365 days. The actual difference is about 11 days, not 12 days. Hence, the reason is incorrect.

26. Correct Answer : A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Explanation: The sidereal year is slightly longer than the tropical year due to the precession or wobbling of Earth's axis. Since the Gregorian calendar is based on the tropical year, festivals based on the sidereal year slowly shift in the Gregorian calendar over time.

MATRIX MATCHING TYPE

27. (a)–(q), (b)–(t), (c)–(r), (d)–(s)

Explanation:

(a) Dakshinayanam refers to the apparent southward shift of the Sun.

(b) A Hybrid Solar Eclipse appears total at some places and annular at others.

(c) Adhika Maasa is an extra month added in certain years in a luni–solar calendar.

(d) A Penumbral Lunar Eclipse occurs when the Moon passes through Earth's outer shadow.

28. (a)–(q), (b)–(p), (c)–(r), (d)–(s)

Explanation:

(a) Titan is known for its thick, cloudy atmosphere.

(b) Ganymede is the largest moon in the solar system.

(c) Triton shows evidence of ice volcanoes.

(d) Miranda appears geologically broken and reformed due to past tectonic activity.

COMPREHENSION TYPE

29. Correct Answer : B) It reflects sunlight

Explanation: The Moon does not produce its own light. It appears bright because it reflects sunlight falling on its surface.

30. Correct Answer : C) The illuminated half faces away from Earth

Explanation: A new Moon occurs when the sunlit half of the Moon faces away from Earth, so it is not visible from Earth.

31. Correct Answer : B) Moon's revolution and changing position relative to Earth and Sun

Explanation: The phases of the Moon occur due to its revolution around Earth and the changing angle between the Sun, Earth, and Moon. The Moon's actual shape does not change.

32. Correct Answer : B) 11 days

Explanation: A solar year is about 365 days and a lunar year is about 354 days. Difference = 365 − 354 = 11 days approximately.

33. Correct Answer : C) Adjust for the extra quarter day in a solar year

Explanation: A solar year is about 365.25 days. The extra 0.25 day each year accumulates to 1 day in 4 years, so a leap day is added to maintain seasonal alignment.

34. Correct Answer : B) Align lunar months with the solar year

Explanation: Adhika Maasa is added in a luni–solar calendar to adjust the 11-day yearly difference and keep lunar months aligned with solar seasons.

INTEGER TYPE

35. Correct Answer : 66

Explanation: Difference per year = 365 − 354 = 11 days.
In 6 years = 11 × 6 = 66 days.

36. Correct Answer : 6

Explanation: 10 hours = 10 × 60 = 600 minutes.
Number of orbits = 600 ÷ 100 = 6 complete orbits.

PREVIOUS YEAR QUESTIONS

1. Correct Answer : C. I, III and V only

Explanation: Statement I is correct because Venus appears very bright and can be seen just before sunrise and just after sunset, so it is called the Morning Star and Evening Star.
Statement II is incorrect because Venus shows retrograde rotation. It rotates from east to west, so the Sun rises in the west and sets in the east on Venus.
Statement III is correct because Venus is nearly equal to Earth in size, mass and density, so it is called Earth's twin.
Statement IV is incorrect because Venus is the second planet from the Sun and the nearest planet to Earth, not the third nearest to the Sun.
Statement V is correct because Venus has a thick carbon dioxide atmosphere that causes a strong greenhouse effect, making it the hottest planet in the solar system.

2. Correct Answer : (a) - (iv), (b) - (i), (c) - (ii), (d) - (v), (e) - (iii)

Explanation: Jupiter spins very rapidly, completing one rotation in about 10 hours, so a day is less than 10 hours.
Saturn has an average density less than that of water, so it would float in water if a large ocean existed.
Mars is called the Red Planet because of iron oxide (rust) on its surface.
Neptune has fourteen known moons.
Venus rotates very slowly and in the opposite direction compared to most planets, which is called retrograde rotation.

3. Correct Answer : C. The pole star appears to be stationary from the earth.

Explanation:
Stars appear to move from east to west due to Earth's rotation from west to east, so option A is incorrect.
The solar system has eight planets because Pluto is classified as a dwarf planet, so option B is incorrect.
The Pole Star lies almost along the Earth's axis of rotation. Therefore, as Earth rotates, it appears nearly fixed in the sky.
Earth revolves around the Sun in an elliptical orbit according to Kepler's First Law, not a perfect circle, so option D is incorrect.

4. Correct Answer : B. I and II only

Explanation:
Artificial satellites are placed in orbits much closer to Earth compared to the Moon, which is about 384,400 km away.
They are used for weather forecasting, remote sensing, navigation and long-distance communication.
Statement III is incorrect because artificial satellites are widely used for telecommunication purposes.

5. Correct Answer : B - SEM SME

Explanation: During a Full Moon, the Earth (E) is positioned between the Sun (S) and the Moon (M), making the alignment S-E-M (or M-E-S). During a New Moon, the Moon (M) is positioned between the Sun (S) and the Earth (E), making the alignment S-M-E (or E-M-S). Option B correctly reflects these geometric alignments.

6. Correct Answer : (d) Venus

Explanation:
Venus is the brightest planet visible from Earth. Because it orbits closer to the Sun than Earth, it appears near the Sun in the sky and is visible either before sunrise or after sunset.
Therefore, it is called the Morning Star and Evening Star.